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Prelubrication Device for Heavy-Duty and High-Horsepower Engines

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Engines discussed in this bulletin are the L10, M11, N14, V28, K19, QSK19, QST30, K38, QSK45, K50, QSK60, and QSK78 engines. This bulletin guides users through the proper installation, usage, and troubleshooting of the prelubrication system.

The Cummins® PRELUB™ system is designed to prelubricate the engine before start-up. Usage of the system will:

- Reduce the risk of a dry start-up
- Prefill new lubricating oil filters at oil change
- Reduce wear of pressurized friction surfaces due to pressure delays after start-up.

There are four models of the PRELUB™ system; Integral, Remote, High Volume Pump (HVP) and Compact.

- The Integral-Mounted has the pump manufactured with a specially designed starter motor on a Delco Remy® 50MT base.
- The Remote-Mounted has a remote bracket that mounts the pump and motor as a self-contained package.
- The HVP and Compact pumps can be either engine or chassis mounted
- There also exists a Periodic PRELUB™ System for QSK60 MCRS generator sets which uses the Compact PRELUB™ pump.

System and Component Description

⚠ CAUTION ⚠

The timer solenoid canister must be mounted with the canister horizontal and the axis of the canister at 90 degrees to the crankshaft centerline. Mounting the canister horizontally will minimize the chance of the switch being accidentally actuated because of shock loading (up and down motion) generated by rough terrain. Mounting the canister at 90 degrees to the crankshaft centerline will minimize the chance of the switch being actuated due to quick starting and stopping. Additionally, the unit must not be mounted on the engine or near the exhaust components.

Note : The lower starter location **must** be used to reduce the amount of lift required by the prelubrication pump in order to reduce the risk of dry or delayed pumping.

Integral System Components

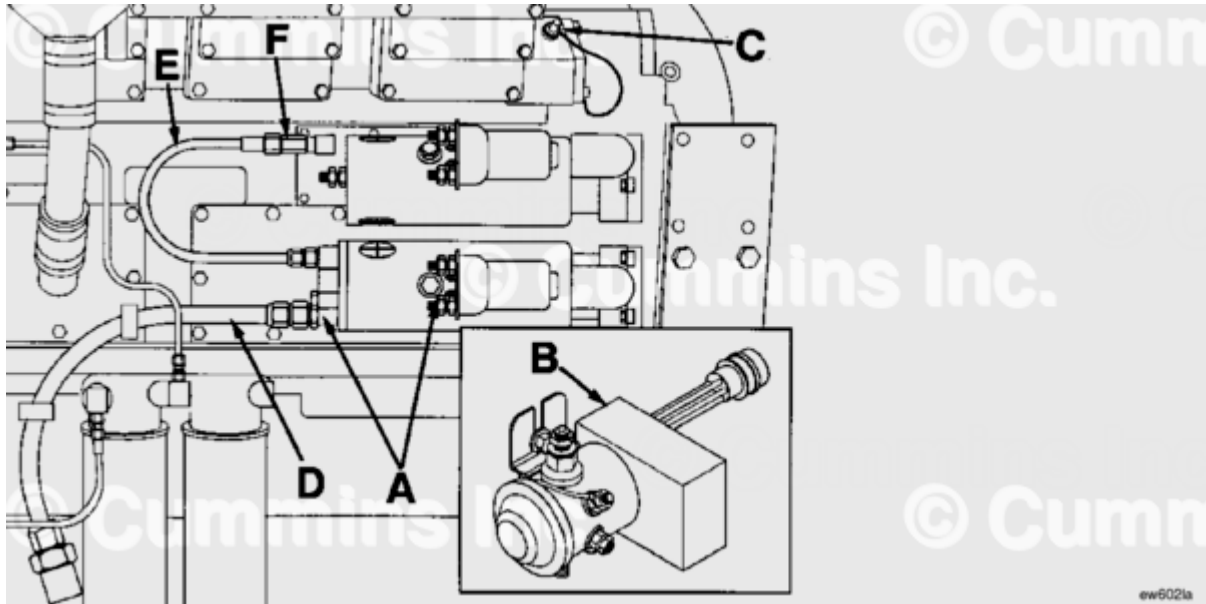


Figure 1, Integral Dual System K38/K50 left bank

- Starter and Pump Assembly
- Timer Solenoid (Horizontally Mounted and 90 Degrees to Crankshaft Centerline)
- Oil Pressure Switch
- Suction Line
- Outlet Line
- Check Valve.

*Wiring harness and miscellaneous hardware (**not** shown). Reference the Integral Dual System graphics above.

Integral Systems can be applied in applications where electric starters are used. The prelubrication starter/pump can then be substituted for the lower conventional starter.

Remote System Components

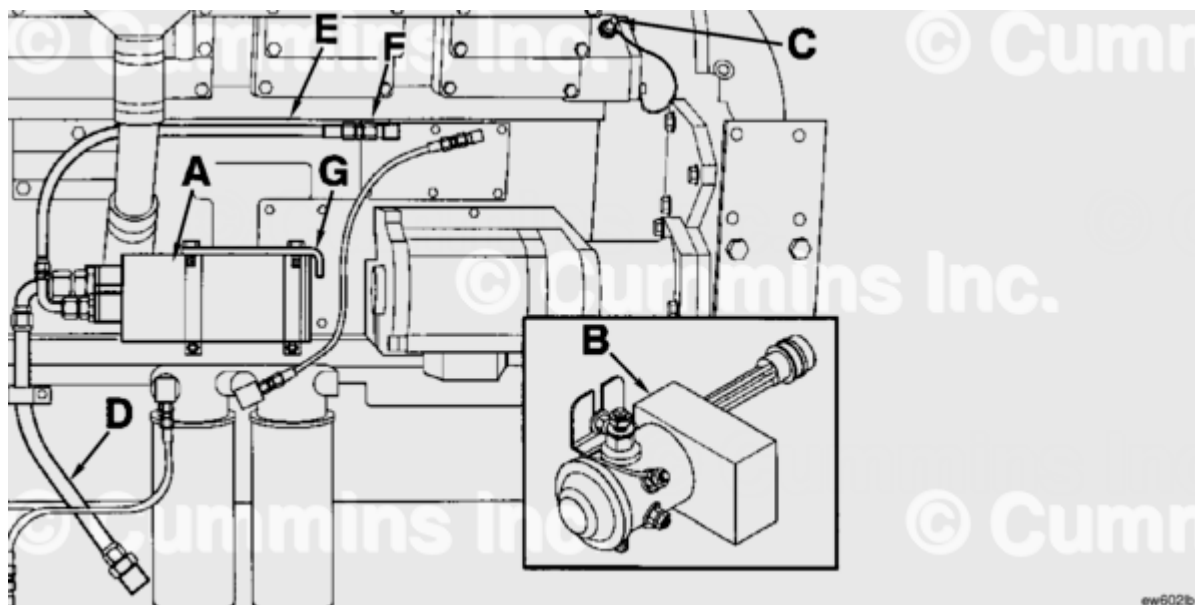
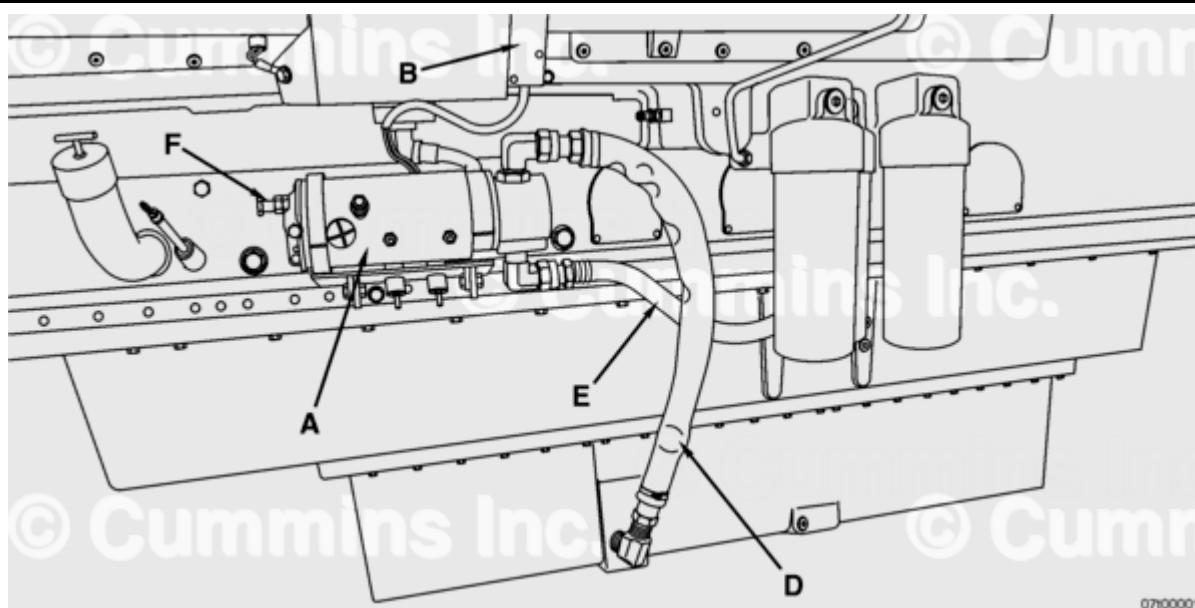


Figure 2, Remote Starter System K38/K50 left bank

- Motor and Pump Assembly
- Timer Solenoid (Horizontally Mounted and 90 Degrees to Crankshaft Centerline)
- Pressure Switch
- Suction Line
- Pressure Line
- Check Valve
- Remote Mounting Bracket.

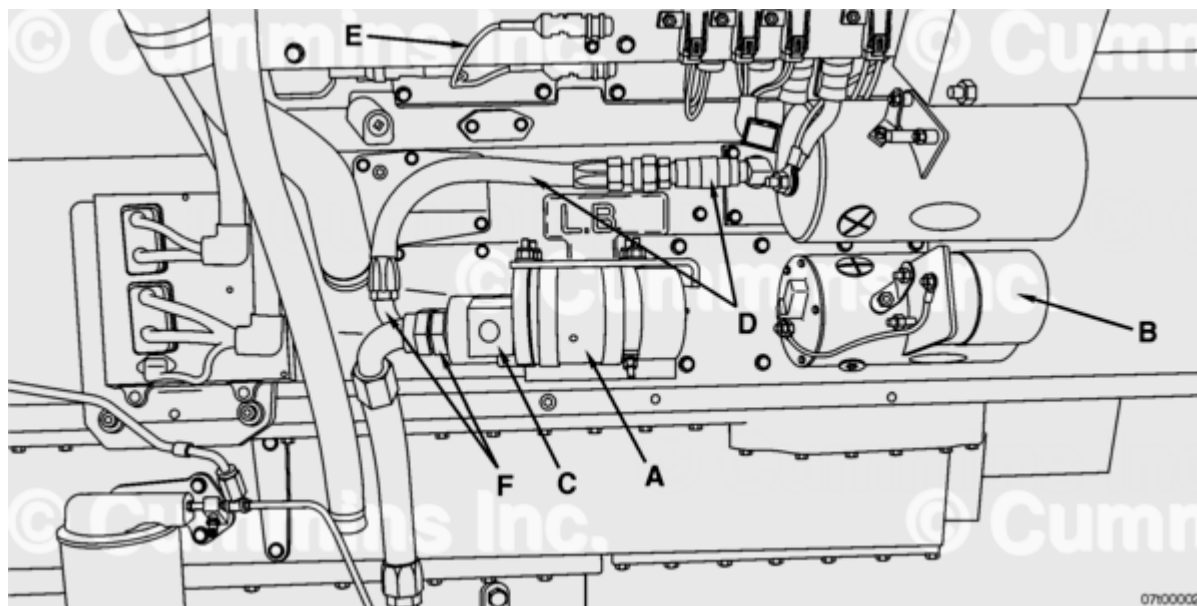
*Wiring harness and miscellaneous hardware (not shown). Reference the Remote Starter System graphics above.



HVP PRELUB™ System

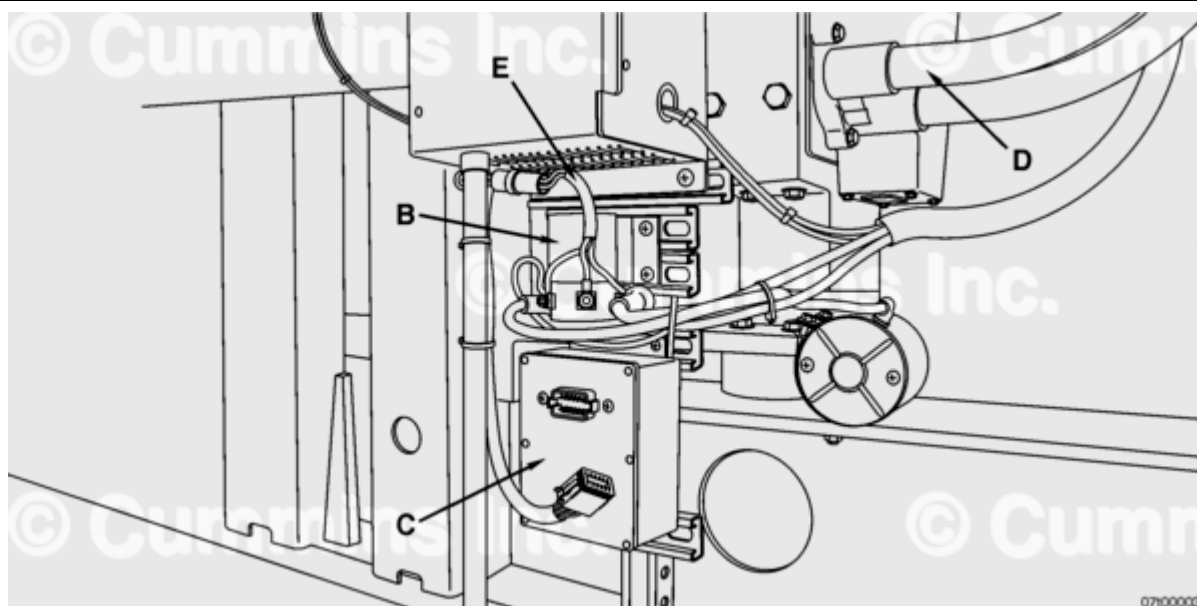
- A. Motor and Pump Assembly
- B. Timer Solenoid (horizontally mounted and 90 degrees to crankshaft centerline)
- C. Pressure switch

- D. Suction Line
- E. Pressure Line
- F. Check Valve



Compact PRELUB™ System

- A. Prelubrication Pump Assembly
- B. Relay and Magnetic Switch
- C. Pressure Switch
- D. Plumbing
- E. Wiring and Harness With Fuse
- F. Oil Suction Ports



Periodic PRELUB™ System

Note : Magnetic switch and Periodic PRELUB™ module should be mounted off engine to reduce vibration exposure.

- A. Prelubrication Pump Assembly
- B. Magnetic Switch
- C. Timer Module
- D. Plumbing
- E. Wiring and Harness With Fuse
- F. Oil Suction Ports

Note : Remote systems are typically used when air starters are present. This arrangement allows the prelubrication pump/motor to be mounted in a remote location on the engine or vehicle chassis.

Cummins® Parts Distribution Center Part Numbers

Field retrofit kits and service replacement components are presently available from Cummins® Parts Distribution Center and Diesel ReCon®. Parts lists are shown below.

K19 and V28 Cummins® Distribution Center Part Numbers - Prelubrication System

Part Description	K19 Integral Part Number	K19 Remote Part Number	V28 Integral Part Number	V28 Remote Part Number
Assembly, Motor and Pump	3084198	368965	3084198	3628965
Switch, Oil Pressure	4984591	4984591	4984591	1984591
Harness, Wiring	3084199	3628758	3084199	3628758
Valve, Check	3628751	3628751	3628751	3628751
Solenoid, Timer	3628971	3628971	3628971	3628971

QSK19 Cummins® Distribution Center Part Numbers - Prelubrication System

Part Description	Right Bank - Integral
Assembly, Motor and Pump	3084198
Switch, Oil Pressure	4984664
Harness, Wiring	3084199
Valve, Check	3084202
Solenoid, Timer	3628971

QST30 Cummins® Distribution Center Part Numbers - Prelubrication System

Name	Integral Part Number		Compact Pump Part Number
Part Description	Left Bank	Right Bank	Compact Pump

QST30 Cummins® Distribution Center Part Numbers - Prelubrication System			
Name	Integral Part Number		Compact Pump Part Number
Assembly, Motor and Pump	3628757	3628997	4295505
Switch, Oil Pressure	4984789	4984789	4924925
Harness, Wiring	3084199	3092612	3643003
Valve, Check	3092150	3092150	3092150
Solenoid, Timer	3628971	3628971	N/A
Switch, Relay	N/A	N/A	4951860
Switch, Magnetic	N/A	N/A	4951864

K38 and K50 Cummins® Distribution Center Part Numbers - Prelubrication System		
Part Description Either Bank	Integral Part Number	Remote Part Number
Primer, Engine Lubrication	3084198	3628965
Switch, Oil Pressure	3628750	3628750
Harness, Wiring	3084199	3628758
Valve, Check	3633246	3633246
Solenoid, Timer	3628971	3628971

QSK38 and QSK50 Cummins® Distribution Center Part Numbers - Prelubrication System		
Part Description Either Bank	Integral Part Number	Remote Part Number
Primer, Engine Lubrication	3084198	3628965
Switch, Oil Pressure	3628750	3628750
Harness, Wiring	3084199	3628758
Valve, Check	3633246	3633246
Solenoid, Timer	3628971	3628971

QSK38 and QSK50 Cummins® Distribution Center Part Numbers - Prelubrication System	
Part Description	Compact Part Number
Primer, Engine Lubrication	3637969
Switch, Oil Pressure	4924925
Harness, Wiring	3643003
Valve, Check	3643839
Relay	4951860
Switch, Magnetic	4951864

QSK38 and QSK50 Marine Cummins® Distribution Center Part Numbers - Prelubrication System	
Part Description	Compact Part Number
Assembly, Motor and Pump	4995212
Relay, Magnetic Switch	N/A
Switch, Oil Pressure	4924925
Harness, Wiring	5254229
Valve, Check	3643839
Bracket, Pump	3639540, 3639541, 4994742

QSK45 and QSK60 Cummins® Distribution Center Part Numbers				
Part Description	Integral LB	Integral RB	Remote	Compact - Side ports plugged
Assembly, Motor and Pump	3410641	3171100	4086437	3637969
Switch, Oil Pressure	4984638	4984638	4101287	4924925
Harness, Wiring	3628758	3628758	4065275	3643003
Valve, Check	4015977	4015977	3643839	3643839
Solenoid, Timer	3628971	3628971	4065468	N/A
Relay	N/A	N/A	N/A	4951860
Switch, Magnetic	N/A	N/A	N/A	4951864

QSK60 Cummins Distribution Center Part Numbers - Compact Prelubrication System (End Ports Plugged)	
Part Description	Part Number
Assembly, Motor and Pump	3639044
Switch, Oil Pressure	4924925
Harness, Wiring	3643003
Valve, Check	3643839
Bracket, Pump	3639540, 3639541
Relay	4951860
Switch, Magnetic	4951864

QSK60 Gas Cummins® Distribution Center Part Numbers	
Part Description	Part Number
Assembly, Motor and Pump	2886099
Switch, Oil Pressure	N/A
Harness, Wiring	3643003
Valve, Check	3643839
Bracket, Pump	3637970
Relay	4951860
Switch, Magnetic	4951864

QSK78 Cummins® Distribution Center Part Numbers			
Part Description	Remote Part Number	Compact - End Ports Plugged	Compact - Side Ports Plugged
Primer, Engine Lubrication	4007201	3639044	3637969
Switch, Oil Pressure	4101287	4924925	4924925
Harness, Wiring	4065275	3643003	3643003
Valve, Check	3643839	36438839	3643839
Solenoid, Timer	4065468	N/A	N/A
Relay	N/A	4951860	4951860
Switch, Magnetic	N/A	4951864	4951864

Operation and Component Function

The Cummins® prelubrication starter is activated by the vehicle's keyswitch or starter button. Turning and holding the key to the START position sends current to the prelubrication solenoid timer, which allows current to the prelubrication starter, and then prelubricates the engine. This solenoid timer prevents current flow to the conventional starting motor solenoid until lubricating oil pressure has been attained in the camshaft oil rifle. After a subsequent 2- to 3-second delay, current is then directed to the vehicle's starting motor, which cranks the engine.

Engine cranking and starting is performed with sufficient lubrication to protect the engine bearings and other components.

HVP PRELUB™ System

The HVP PRELUB™ system is activated when the operator turns the keyswitch to the start position. With the key in the start position, the prelubrication timer relay is energized to provide power to the pump assembly. The starter solenoid remains de-energized to prevent cranking. The prelubrication pump will draw oil from the oil pan, pass through the oil filters, and fill the oil galleries of the engine. Once the main oil rifle is pressurized, a normally closed switch will open. The timer solenoid recognizes the open circuit at the pressure switch and de-energizes the time solenoid relay to stop the motor.

The timer relay will immediately activate the cranking circuit for HVP PRELUB™ systems. The time for the PRELUB™ system to generate sufficient pressure will be affected by oil viscosity. Consequently, for a given oil grade, the time to reach pressure will increase with cold temperatures. In order to maintain acceptable prelubrication times, apply arctic weather treatments as advised in the engine Operation and Maintenance Manual.

Compact PRELUB™ System

The Compact PRELUB™ pump is activated when the operator turns the keyswitch to the start position. With the key in the start position, the prelubrication relay and magnetic switch are energized to provide power to the pump assembly. The prelubrication pump will draw oil from the oil pan, pass through the oil filters, and fill the oil galleries of the engine. Once the main oil rifle is pressurized, a normally open pressure switch will close. The relay recognizes the closed circuit at the pressure switch and de-energizes the magnetic switch to stop the motor. The relay will immediately activate the cranking circuit.

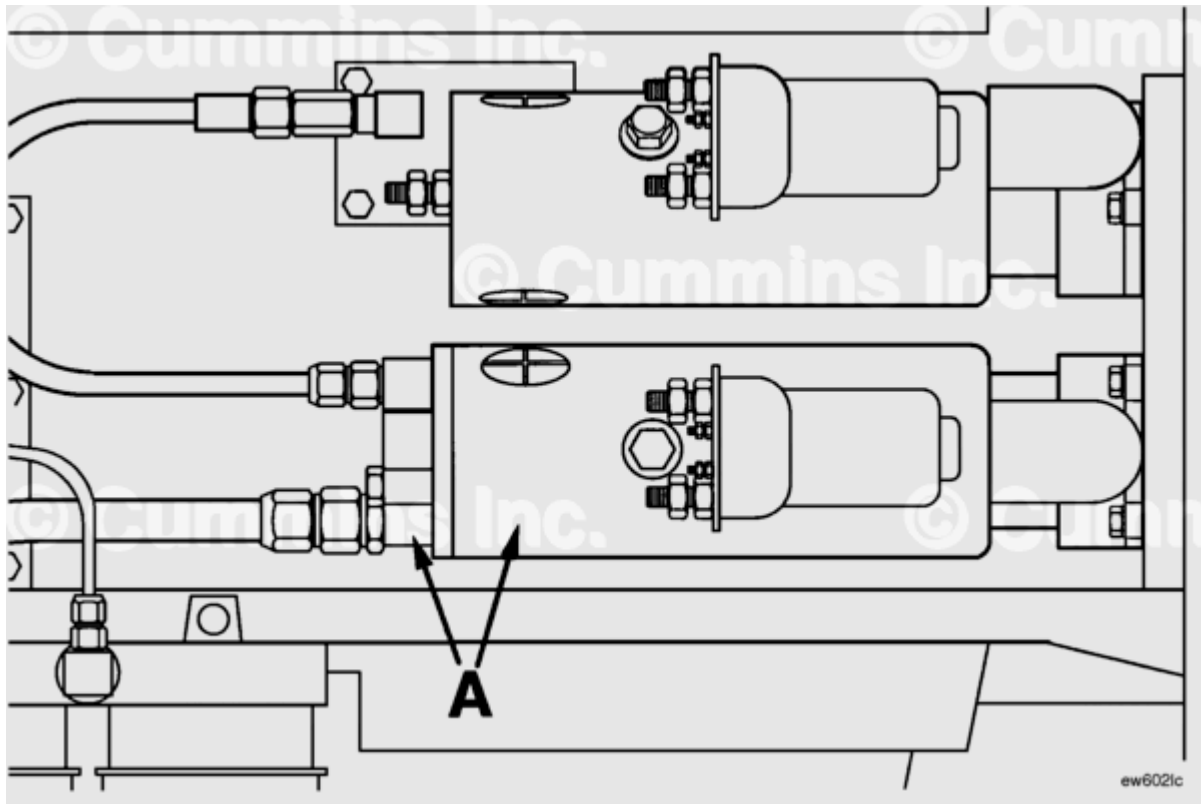
The time required for the prelubrication system to generate sufficient pressure will be affected by oil viscosity. For the Compact PRELUB™ pumps, the maximum duty cycle is a total operation time **not** to exceed 3 minutes ON in a 60 minute time period. The ambient temperature range is from -40 to 125° C (-40 to 257° F). For reference, an electric starter maximum duty cycle is 30 seconds on 120 seconds off.

Periodic PRELUB™ System

- Upon application of power, the module scans the states of the inputs - first the engine cranks signal, then the engine operate signal
- If the engine crank signal is detected, the pump is latched on for 15 seconds (regardless of the time the crank signal is present) then the pump operation is locked out for 10 minutes. Multiple start attempts or periodic cycles will **not** operate the pump during this 10 minute period.
- If the engine operate signal is present, the pump will **not** operate (or shut off if operating), then the program repeats its loop of scanning the input.
- If neither signal is present, a 2 minute timer begins. Inputs are repeatedly scanned during the 2 minutes as well - if either is detected, operation listed in steps 2 or 3 above occurs. If engine was started and operated less than 10 minutes, the 10 minute lockout counter **must** finish timing before the 2 minute timer begins or another crank-initiated cycle operates

- At the end of the 2 minute count, the pump is switched on for 40 seconds. Again, inputs are repeatedly scanned during the 40 second operation.
- After the 40 second operation, the pump is switched off for 12 hours. Inputs are scanned during this rest period.
- The program continues this loop until power is removed.

Installation Guidelines



Motor and Pump Assembly (A)

Retrofit installations of Cummins® prelubrication systems typically require 2 to 6 hours. The prelubrication starter is installed in the lower starter bore for integral systems if the engine has two starters. If the engine has one starter, the PRELUB™ starter replaces the conventional starter. Remote systems require an additional bracket for mounting the prelubrication motor in an available location no higher than the lower starter bore in the flywheel housing, no farther than 1270 mm [50 in] from the suction port in the lubricating oil pan. In remote installations, the unit is **not** to be mounted higher than 85 mm [3.35 in] below the crankshaft centerline.

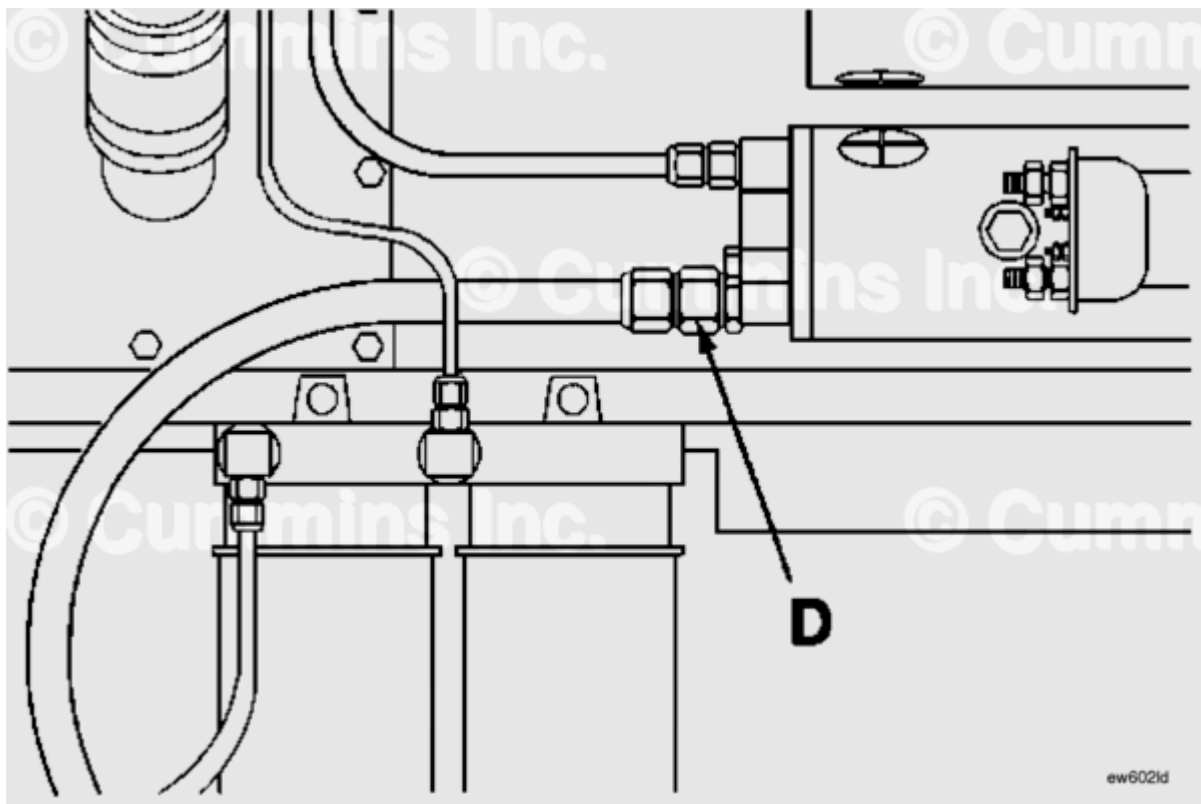
Engine Mounted Compact Pump

Engine mounted Compact PRELUB™ systems are utilized on the 30L, 38L, 50L, some 60L, and 78L engines. The prelubrication pump/motor, hoses, and pressure switch are mounted at the factory. The QSK45/60/78 engines have metric bolts to mount the pump. The K38/50 and the QSK38/50 engine have imperial bolts to mount the pump.

Chassis Mounted Compact PRELUB™ Pump

Chassis mounted Compact PRELUB™ systems are utilized on 45L and some 60L engines. These arrangements require a mounting plate for the prelubrication motor. Mounting straps are supplied in a kit with the engine. The mounting bolts are included in the kit. This includes (1) M8 nut and bolt, and (2) M12x1.75 bolts that are 30mm [1.18 in] long.

Suction Line (D)

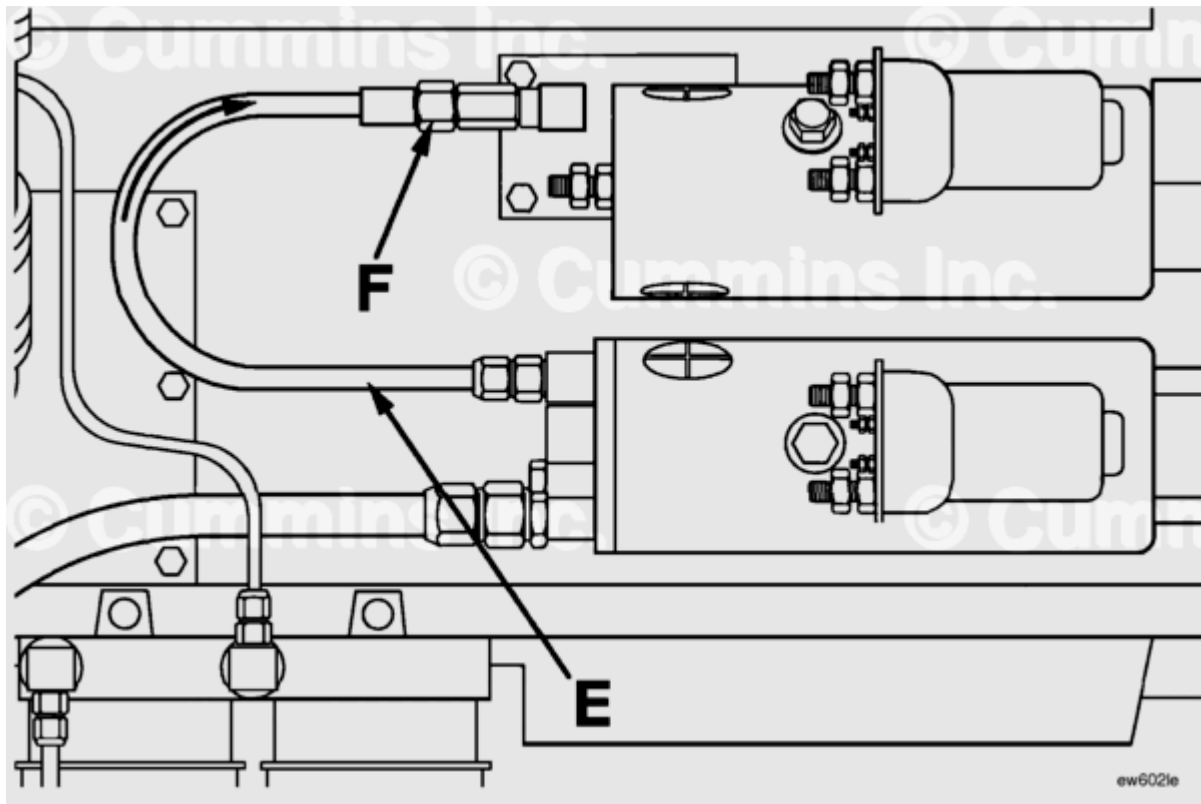


The large suction hose connects the lubricating oil pan sump to the prelubrication pump. An additional port has been added to new K38 and K50 lubricating oil pans for this purpose. The length of the suction side hose should be minimized. The oil supply port location varies by engine model. See the oil pan option in the sales handbook for the designated prelubrication port. Retrofit installations use the lubricating oil heater port nearest the pump. Do **not** use the lubricating oil drain port for retrofit installation purposes. It is recommended that the suction side hose included with the PRELUB™ system is used in the engine installation. However if a custom hose is used, reference the AEB24.20 design guidelines.

Additional recommendations unique to the PRELUB™ system include:

- The suction hose length is not to exceed 1270mm [50in] in length
- The oil pan suction fitting should have at least 25.4 cm [1in] ID (1 5/8" ORB, 1" NPT, M42)
- Clips on the hose need to be used to avoid rubbing or excessive vibration

Outlet Line (E) and Check Valve (F)

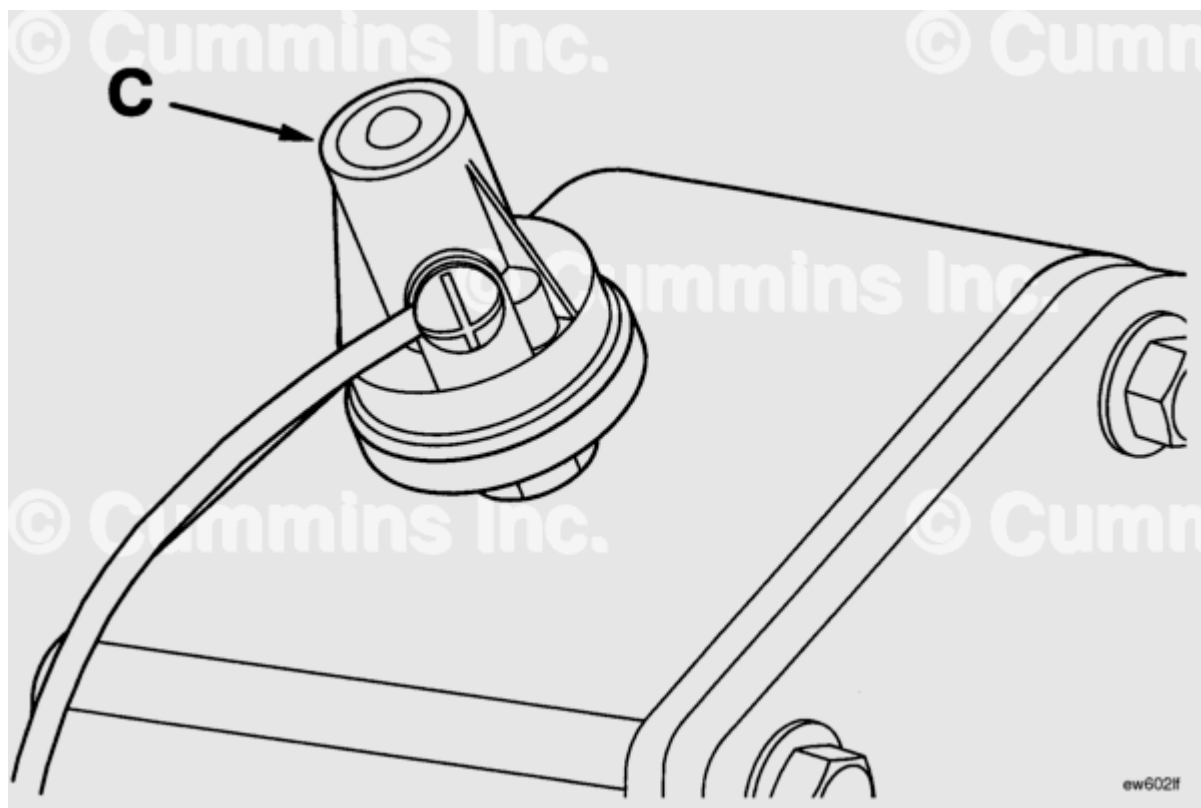


The pressure side hose will connect the prelubrication pump to the engine. The port location for the oil pressure hose will vary by engine model. For every installation, the pressure side hose **must** be connected before the engine lubrication filters. The length of the pressure side hose is limited by the diameter of both the pressure and suction hoses as well as the length of the suction side hose. Installations should use **only** high quality SAE connections and fittings for oil lines.

Maximum hose length: The lesser of 4 times suction hose length or 3658 mm [144 in]. The in PRELUB™ system oil return port should have at least 15.2mm [0.6 in] ID (3/4" pipe, 1 1/16" ORB, M22).

Note : The check valve has an arrow stamped on it indicating flow direction.

The check valve included in the prelubrication kit is to be installed where the pressure side hose is attached to the engine. Flow through the valve is to be toward the engine. The valve will prevent reverse flow after the engine is started. If the check valve is installed incorrectly, prelubrication motor malfunction may result.



Pressure Switch (C)

This switch **must** be located to sense lubricating oil pressure after the lubricating oil has passed through the full flow lubricating oil filters. Typical locations are, the lubricating oil filter head (outboard side), the camshaft covers on the rear of the block, or the side of the block.

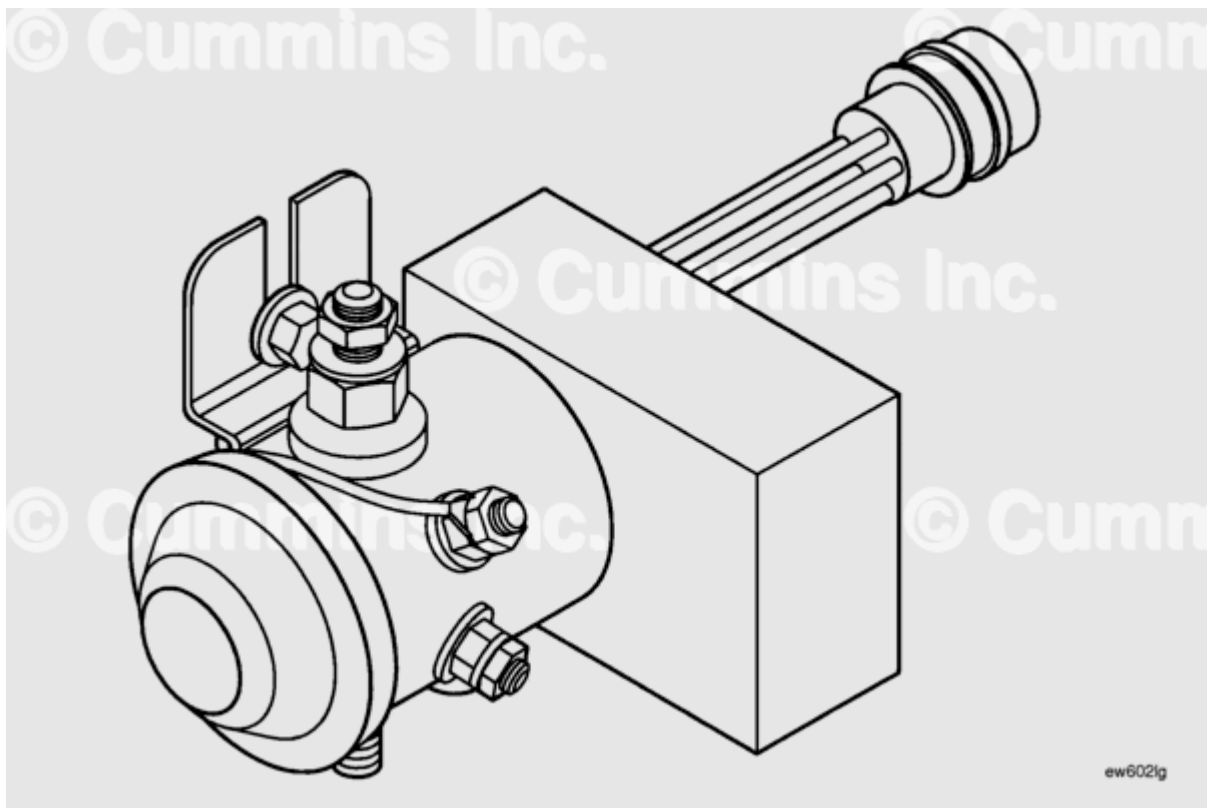
Integral and Remote Systems - A 17 or 31-kPa [2.5 or 4.5-psi] switch (normally closed) is installed on engine.

HVP System - A 31-kPa [4.5-psi] (normally closed) switch is installed on engine

Compact - A 17 or 28-kPa [2.5 or 4.0-psi] (normally open) pressure switch is installed on engine.

Periodic Prelub System - No pressure switch is required due to the inclusion of a timer module.

Relay and Magnetic Switch/Timer Solenoid (B)



⚠ CAUTION ⚠

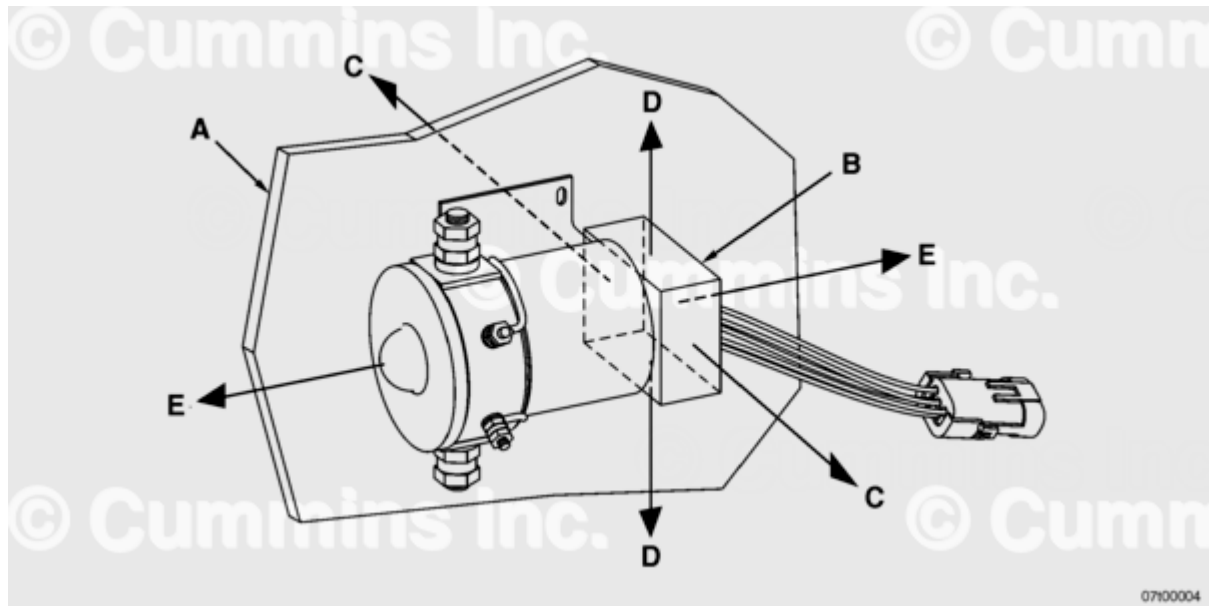
The relay and magnetic switch must be mounted off the engine and away from heat sources such as exhaust components. The timer solenoid/relay canister should be mounted with the canister horizontal and axis of the canister at 90 degrees to the crankshaft centerline. Mounting the canister horizontally will minimize the chance of the switch being actuated because of vertical “shock loading”. Mounting the canister at 90 degrees to the direction of travel will minimize the chance of the switch being activated due to “shock loading” from the vehicle under acceleration or braking.

⚠ CAUTION ⚠

Do not connect a jumper wire across the solenoid. Jumping the solenoid can cause solenoid malfunction.

The timer solenoid controls the prelubrication cycle. Current is supplied to the timer through the keyswitch. The ground path is completed by the normally closed pressure switch that is preset to open at 17 or 28-kPa [2.5 or 4.0-psi]. At switch opening, current is redirected to the vehicle solenoid for engine cranking, following a 2- to 3-second delay. This device is to be mounted off the engine to limit vibration and heat exposure. A chassis location needs to be selected that provides future accessibility without excessive exposure to heat or mechanical damage. The timer will function every starting cycle. Do **not** install a bypass switch; it will make the system discretionary.

Relay and Magnetic Switch



The magnetic switch and relay control the prelubrication cycle. Current is supplied to the relay from the keyswitch which energizes the magnetic switch. The normally open pressure switch energizes the relay once the pressure reaches 4psi switching power from the prelubrication circuit to the starter and cranking the engine. This device is to be mounted off the engine to limit vibration and heat exposure. A chassis location needs to be selected that provides future accessibility without excessive exposure to heat or mechanical damage.

The relay and magnetic switch should **not** be exposed to ambient temperatures over 85°C [185° F]. The relay and magnetic switch should be protected from direct contact of a high pressure wash.

Periodic Control Module

Use 1/4" hardware supplied with the kit to mount the switch and module. The ground wire from the magnetic switch should be wired through the Normally Closed contacts of the relay. System control relay should be operated by the generator set E-Stop (bulk orange wire supplied to make connection). The relay, switch and module contain sensitive electronic components that must be isolated from engine vibration and high heat.

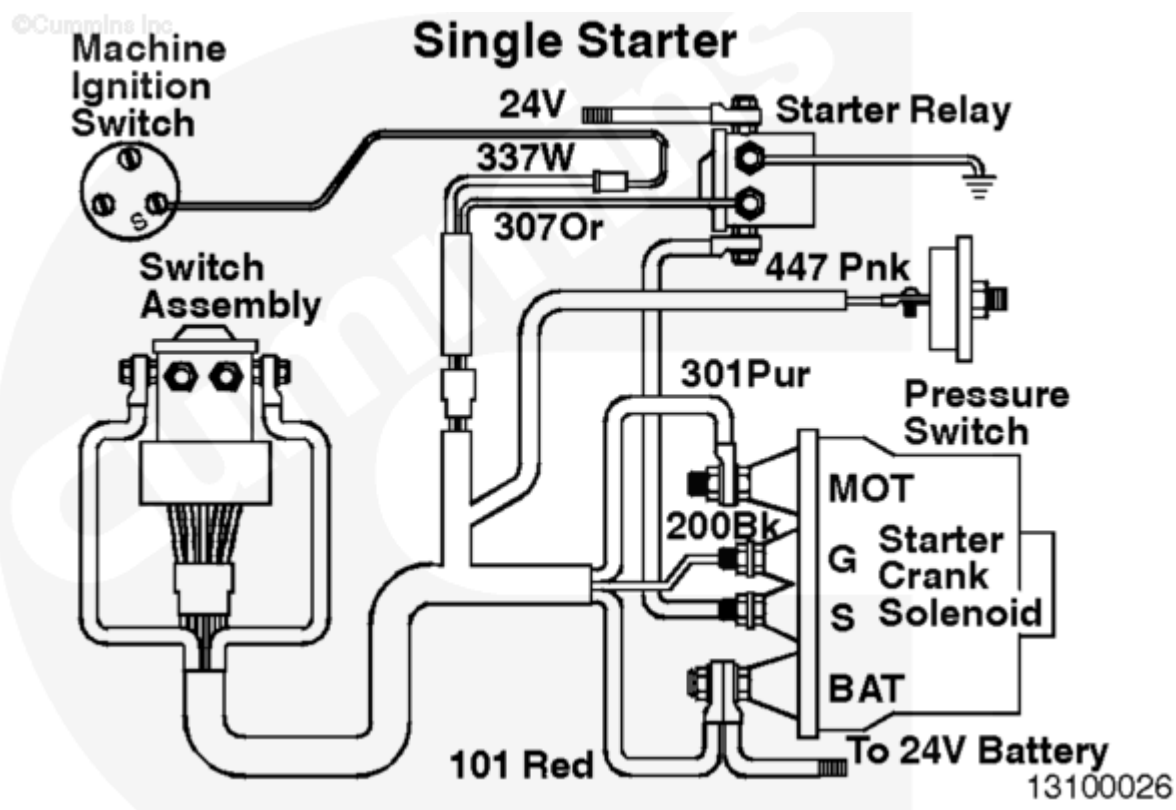


Figure 3, Integral Single-Starter System

Wiring Harness

A wiring harness is supplied with each Cummins® prelubrication system. The wiring schematic illustrates the integral single-starter system, integral dual-starter system, and the remote system.

Note : Integral single-starter systems will not provide adequate cranking for the V28/KV/QSK series engines.

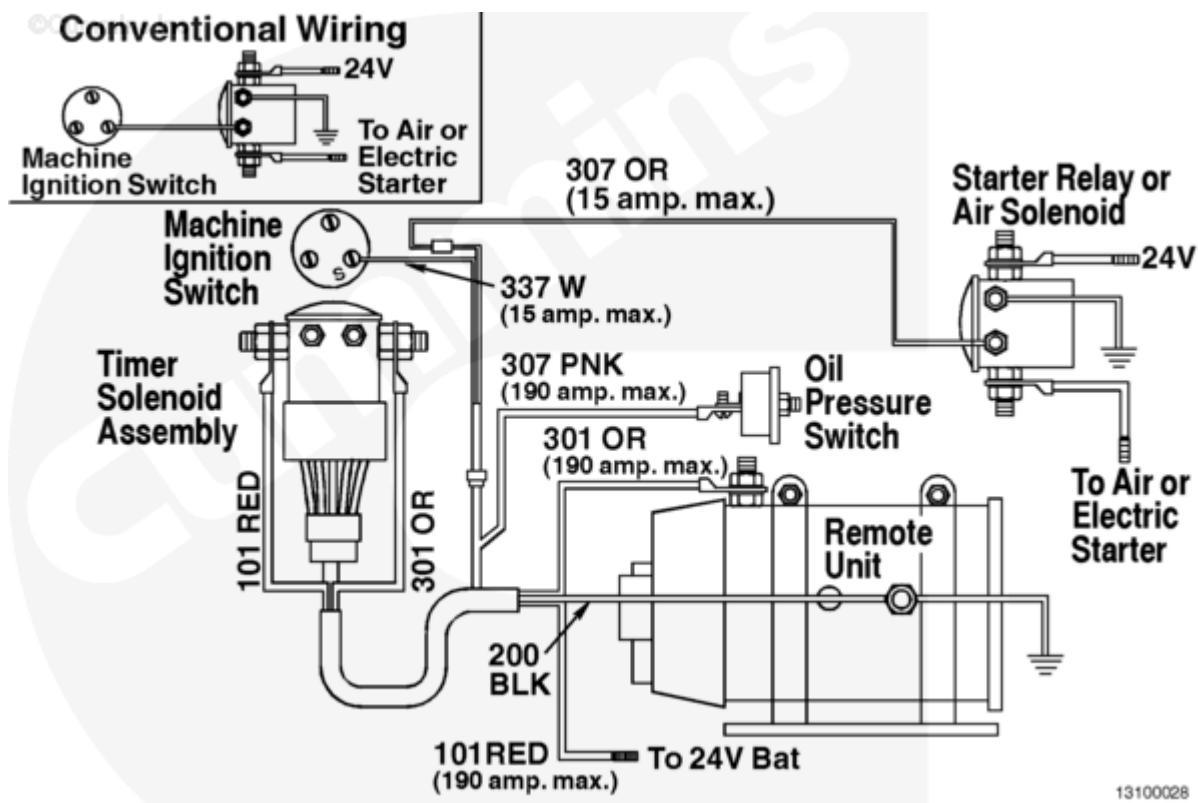
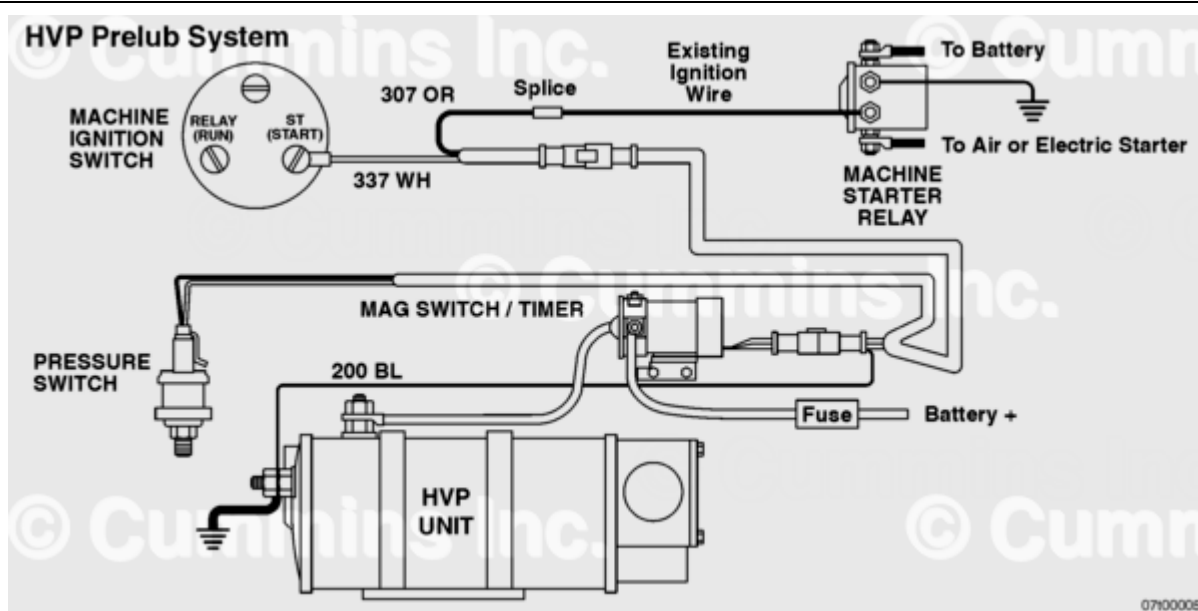
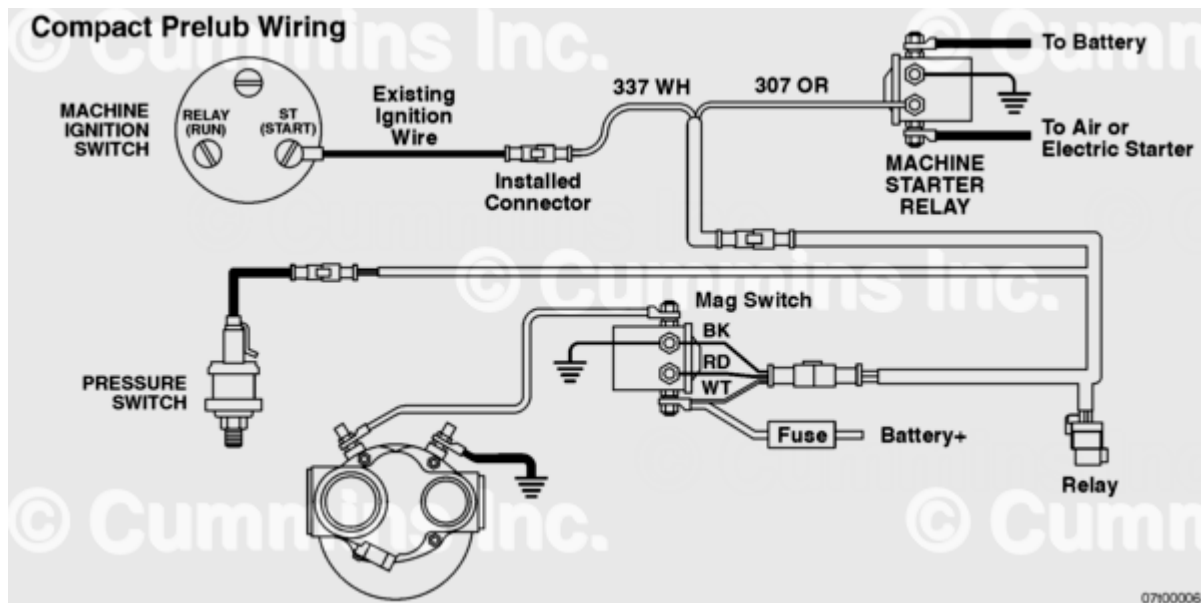


Figure 4.

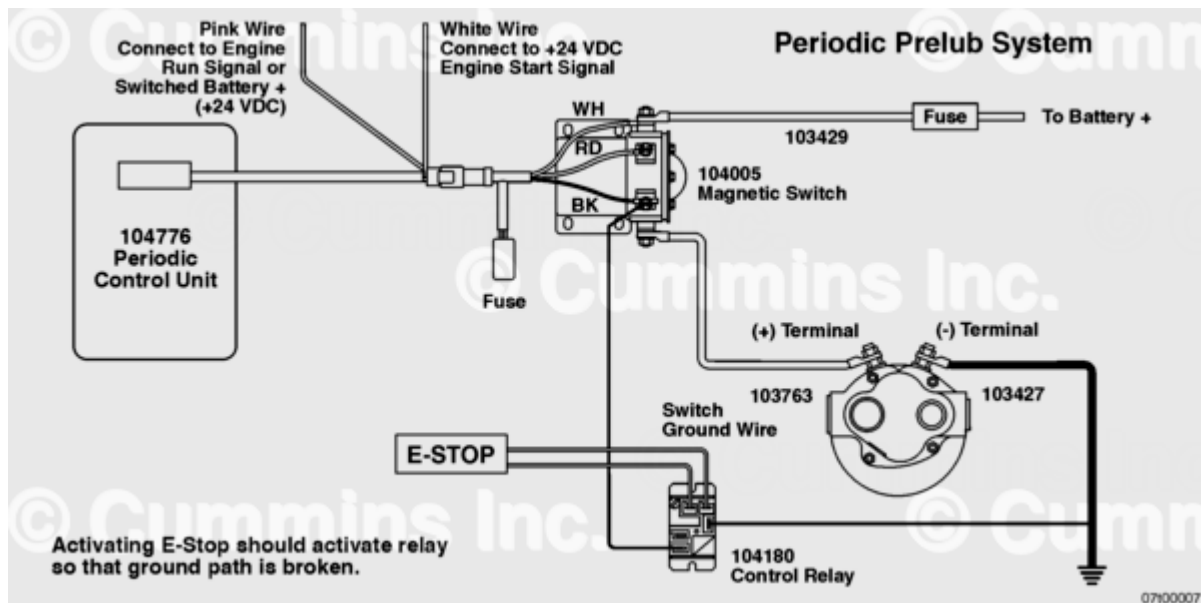
Remote Starter System.



HVP Prelubrication System



Compact Prelubrication System



Periodic Prelubrication System

Engine-Specific Installation and Application Guidelines

The following information contains specific details concerning the usage and application of the prelubrication device of the L10/M11/N14/K19/QSK19/QST30/V28 engines, where different from the K38/K50/QSK45/QSK60/QSK78 engines. Information contained in Section I through Section IV of this bulletin is to be used unless otherwise indicated below.

The primary exceptions are:

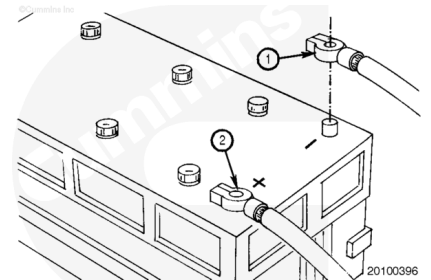
1. The oil pressure switch setting is 28 kPa [4.0 psi], **not** 17 kPa [2.5 psi].

2. The suction hose diameter is a Number 16, **not** a Number 20.
3. The Parts Distribution Center (PDC) numbers and Diesel ReCon® part numbers are different. Reference the previous tables.
4. Location of the pressure switch and plumbing attachment locations for suction and pressure hoses from the pump are different. Reference the following pages.
5. **Only** L10/M11/N14/K19/QSK19/QST30/V28 engines require one starter. Reference the Single-Starter Wiring Diagram graphic.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

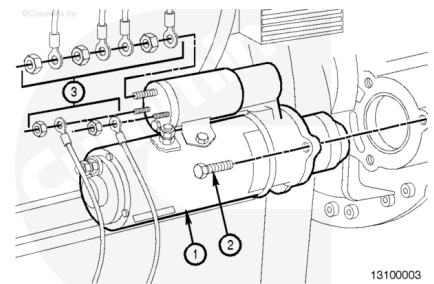
Disconnect the batteries. Refer to the OEM service manual.



L10/M11/N14 Integral PRELUB™ Starter

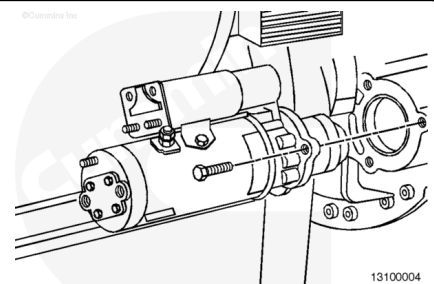
Remove the electrical connections (3) from the starter motor (1), and mark their locations.

Remove the three capscrews (2) and the existing starter motor (1). Save the capscrews.



Install the PRELUB™ starter using the capscrews that were removed above.

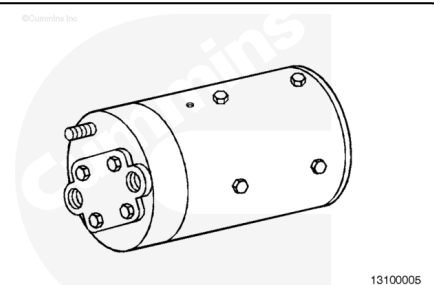
Reference the appropriate Troubleshooting and Repair Manual for the correct torque value.



L10/M11/N14 Remote PRELUB™ Pump

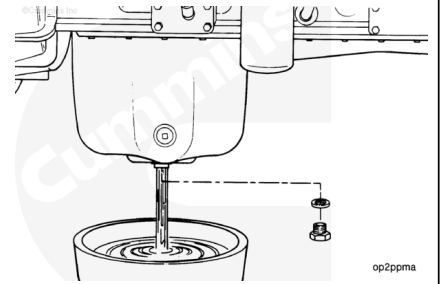
If using a remote pump, the starter does **not** have to be removed or altered.

The remote pump can be mounted in any convenient location on the frame on either side of the engine compartment.



⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and can cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

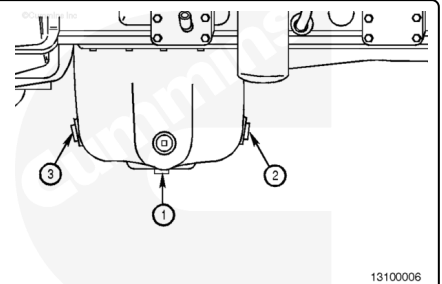


⚠ WARNING ⚠

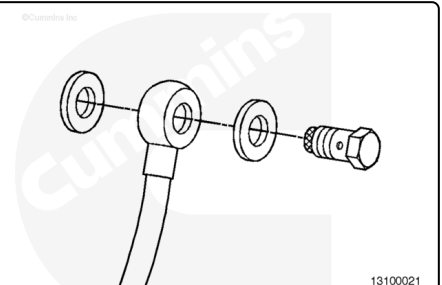
To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

Remove the oil drain plug from the bottom of the lubricating oil pan. Do **not** remove the plugs on either side of the oil pan to drain the oil. They will **not** allow the oil to drain completely.

If an oil change suction device is attached to the side drain outlet (2) on the starter side of the pan, it **must** be moved to the other side drain (3).



Use the banjo fitting provided in the plumbing kit to attach the suction hose to the side drain of the oil pan.

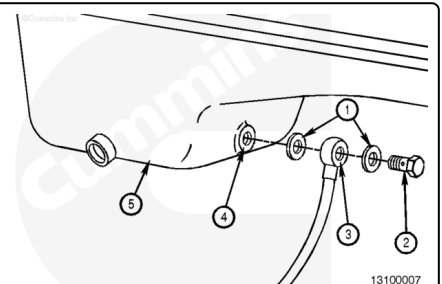


Put one copper sealing washer (1) on the banjo bolt (2).

Insert the banjo bolt into the banjo fitting (3).

Put the other copper sealing washer (1) onto the banjo bolt/banjo fitting assembly.

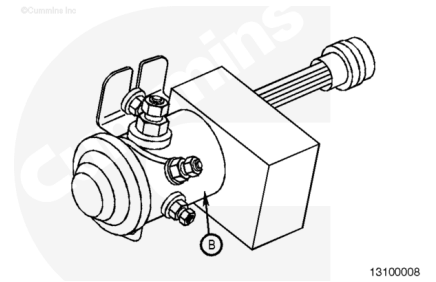
Install the banjo bolt/banjo fitting assembly into the side drain (4) of the oil pan (5).



Torque Value: 88 n•m [65 ft-lb]

⚠ CAUTION ⚠

The timer solenoid canister must be mounted with the canister horizontal and the axis of the canister at 90 degrees to the crankshaft centerline. Mounting the canister horizontally will minimize the chance of the switch being accidentally actuated because of shock loading (up and down motion) generated by rough terrain. Mounting the canister at 90 degrees to the crankshaft centerline will minimize the chance of the switch being actuated due to quick starting and stopping. Additionally, the unit must not be mounted on the engine or near the exhaust components.

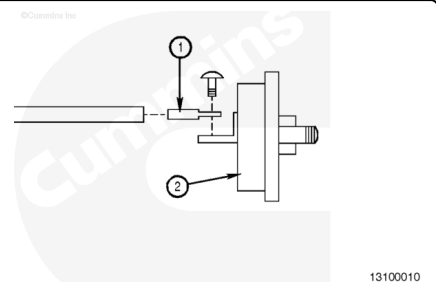


Install the PRELUB™ timer solenoid on the firewall in a convenient location.

Remove the fitting plug located directly above the starter from the main oil rifle.

Install the pressure switch into the fitting of the main oil rifle.

Connect the pink wire (1) in the wiring harness to the pressure switch (2).

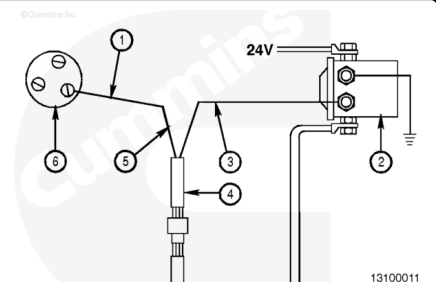


Remove the wire (1) from the starter relay (2) that goes to the starter switch (6).

Cut off the connector.

Splice this wire (1) onto the white wire (5) in the PRELUB™ wiring harness (4).

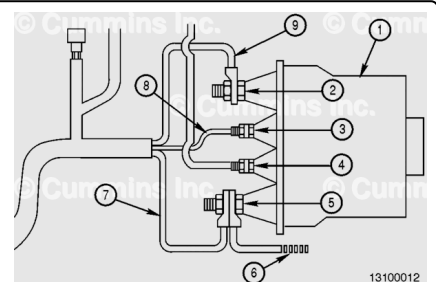
Connect the orange wire (3) to the starter relay terminal from which wire Number 1 was removed.



Integral PRELUB™ Starter Wiring

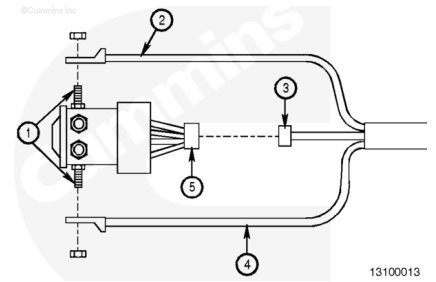
If the old starter had an electrical cutout circuit installed, cut off the harness connector, and splice the wires together.

Install the PRELUB™ starter wiring according to the wiring diagram.



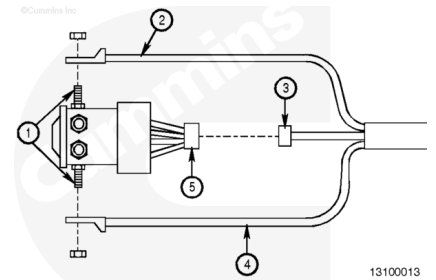
Connect the wiring harness connector (3) to the PRELUB™ timer solenoid connector (5).

Connect the red wire (4) and the purple wire (2) to the timer solenoid terminals (1).



Connect the wiring harness connector (3) to the PRELUB™ timer solenoid connector (5).

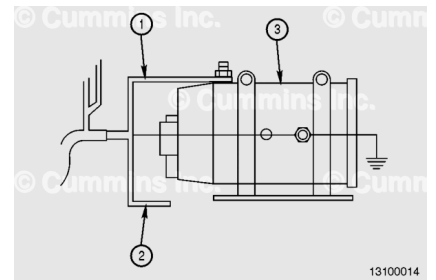
Connect the red wire (4) and the purple wire (2) to the timer solenoid terminals (1).



Remote PRELUB™ Pump Wiring

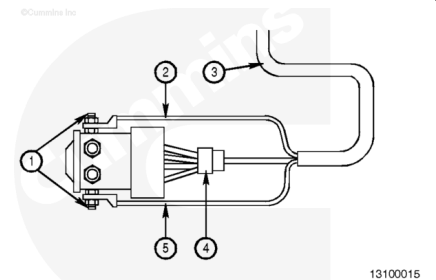
Connect the red wire (1) to the starter terminal.

Connect the orange wire (2) to a 12-VDC or 24-VDC battery connection.



Connect the wiring harness (3) to the timer solenoid connection (4).

Connect the red wire (20) and the orange wire (4) to the timer solenoid terminals (1).

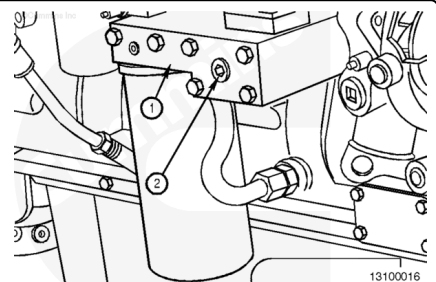


L10/M11 Pressure Line Installation

If an alternate location is used, make sure that it is a dirty oil port.

Remove the plug (2) from the front of the lubricating oil filter head (1).

Note : Location Number 2 is preferred. An alternate location can be used.



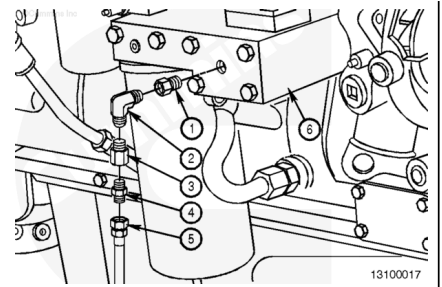
Install a straight thread adapter (1) into the lubricating oil filter head (6) where the plug was removed.

Install a 90-degree elbow (2) into the adapter.

Install a check valve (3) into the elbow. The arrow on the check valve **must** point toward the elbow, in the direction of flow.

Install a male hose adapter (4) into the check valve.

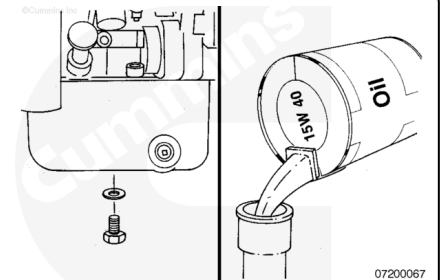
Attach the prelubrication hose (5) to the male hose adapter.



Install the oil drain plug(s) into the lubricating pan.

Torque Value: 88 n·m [65 ft-lb]

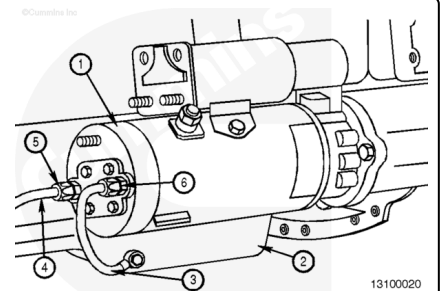
Fill the engine with clean oil to the correct level. Reference the Lubricating Oil System Specifications in the proper Operation and Maintenance Manual for the correct oil capacity.



Fill the suction line (3) with clean engine oil.

Install the suction line from the oil pan (2) to the in port (6) on the PRELUB™ starter (1).

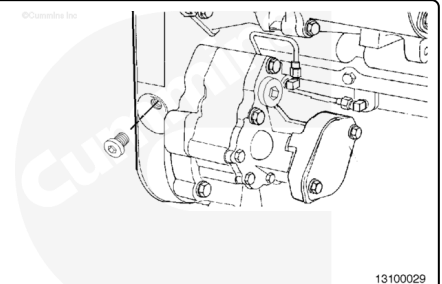
Install the PRELUB™ hose (4) from the lubricating oil filter head to the pressure port (5) on the PRELUB™ starter (1).



N14 Pressure Line Installation

Primary Location

Remove the pressure port plug from the gear housing oil pressure port.



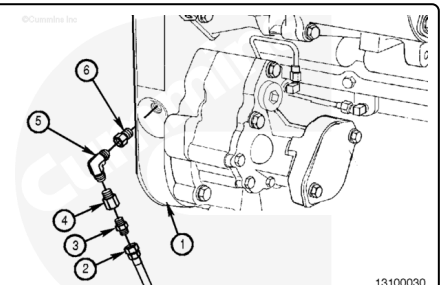
Install a straight thread adapter (6) into the dirty oil port on the gear housing (1).

Install a 90-degree elbow (5) into the straight thread adapter.

Install a check valve (4) into the 90-degree elbow with the arrow pointing toward the elbow, in the direction of flow.

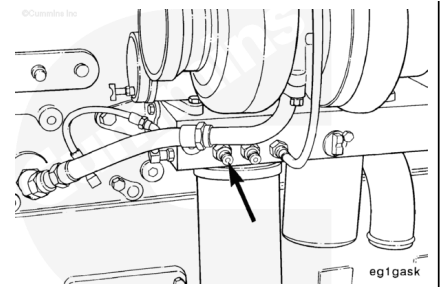
Install a male pipe adapter (3) onto the check valve.

Attach the PRELUB™ hose (2) to the male adapter.



Secondary Location

Remove the pressure port plug (arrow) from the lubricating oil filter head. Alternate locations can be used as long as they are dirty oil ports.



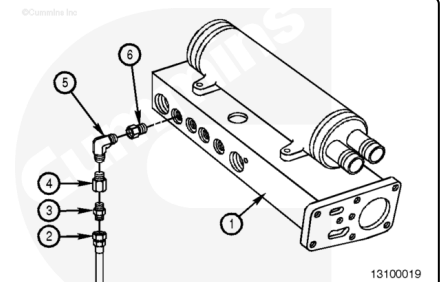
Install a straight thread adapter (6) into the dirty oil port lubricating oil filter head (1).

Install a 90-degree elbow (5) into the straight thread adapter.

Install a check valve (4) into the 90-degree elbow with the arrow pointing toward the elbow, in the direction of flow.

Install a male pipe adapter (3) onto the check valve.

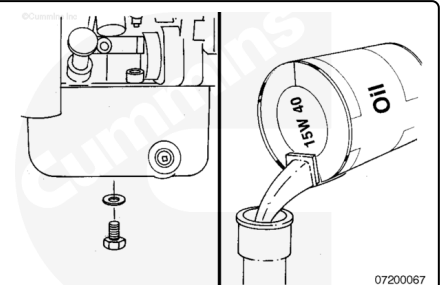
Attach the PRELUB™ hose (2) to the male adapter.



Install the drain plug into the lubricating oil pan.

Torque Value: 88 n·m [65 ft-lb]

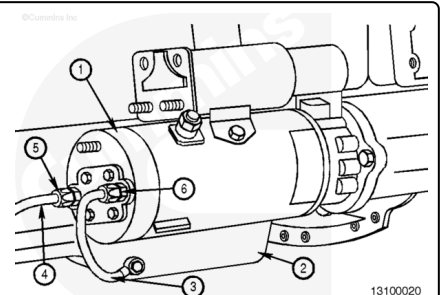
Fill the engine with clean oil to the correct level. Reference the Lubricating Oil System Specifications in the correct Operation and Maintenance Manual for the correct oil capacity.



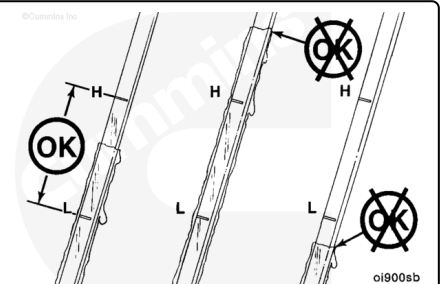
Fill the suction line (3) with clean engine oil.

Install the suction line from the oil pan (2) to the in port (6) on the PRELUB™ starter (1).

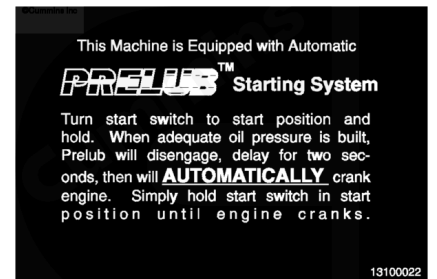
Install the PRELUB™ hose (4) from the lubricating oil filter head to the pressure port (5) on the PRELUB™ starter (1).



Check the oil level on the dipstick. It **must** be filled to between the L (low) and H (high) mark.



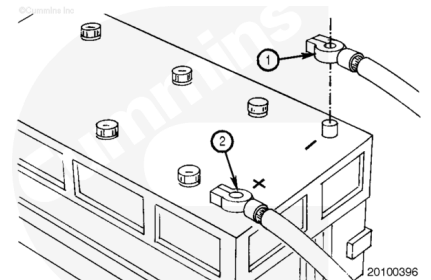
Mount the decal directly below the ignition switch on the vehicle dash panel.



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Connect the battery terminals. Refer to the OEM service manual.

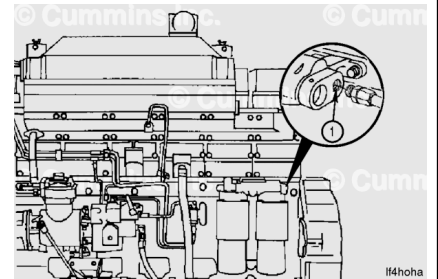


K19 Pressure Hose and Check Valve Installation

⚠ CAUTION ⚠

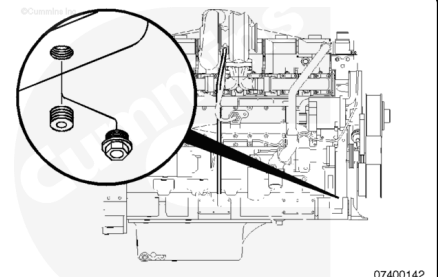
The check valve must be installed with the flow arrow toward the engine. Installing the valve backward will cause the pump to fail.

Use the 3/8-inch American National Standard Taper Pipe Thread (NPT) port in the oil filter head for installing the check valve and pressure hose.



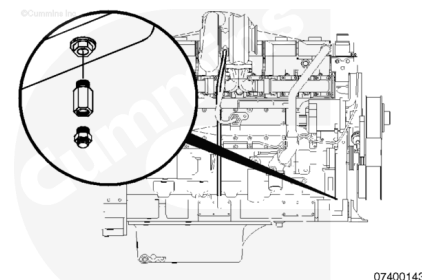
QSK19 Pressure Hose and Check Valve Installation

Locate and remove the plug in the bottom of the lubricating oil pump flange. Replace with a 1 5/8-12 straight thread o-ring x 1/2 American National Standard Taper Pipe Thread (NPT) adapter.



⚠ CAUTION ⚠

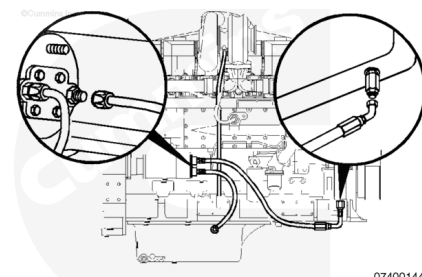
The check valve must be installed with the flow arrow toward the engine. Installing the valve backward will cause the pump to fail.



Install the check valve and Joint Industrial Council (JIC) adapter provided in the plumbing kit.

Note : Use pipe sealant or Teflon® tape when installing the check valve.

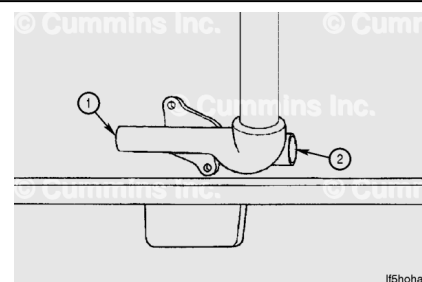
Attach the pressure hose to the adapter. Connect the other end to the OUT port of the PRELUB™ pump. Tighten all connections.



V28 Pressure Hose and Check Valve

⚠ CAUTION ⚠

The check valve must be installed with the flow arrow toward the engine. Installing the valve backward will cause the pump to fail.

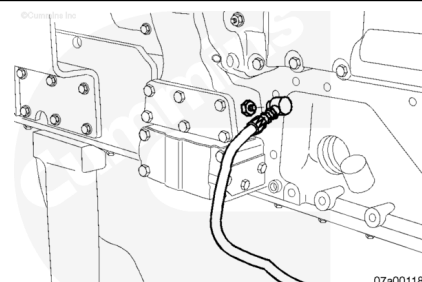


Use the lubricating oil transfer tube assembly below the lubricating oil filter head for installation of the check valve and pressure hose.

QST30 Pressure Hose and Check Valve

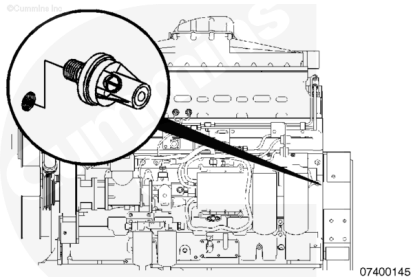
⚠ CAUTION ⚠

The check valve must be installed with the flow arrow toward the engine. Installing the valve backward will cause the pump to fail.



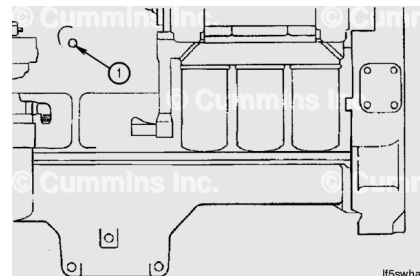
Use the front side of the lubricating oil transfer connection for installation of the check valve and pressure hose.

Install the pressure switch **behind** the oil filters into the main oil rifle on the left bank as shown.



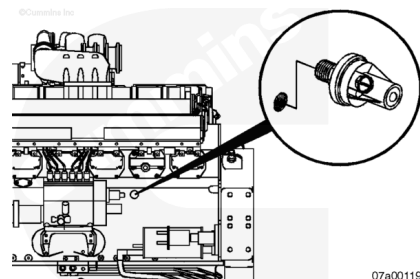
V28 Pressure Switch Installation

Install the pressure switch on the oil filter side of the engine as shown.



QST30 Pressure Switch Installation

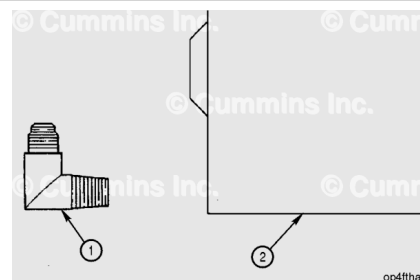
Install the pressure switch on the left bank side of engine after the fuel pump in the center port of the piston cooling gallery as shown.



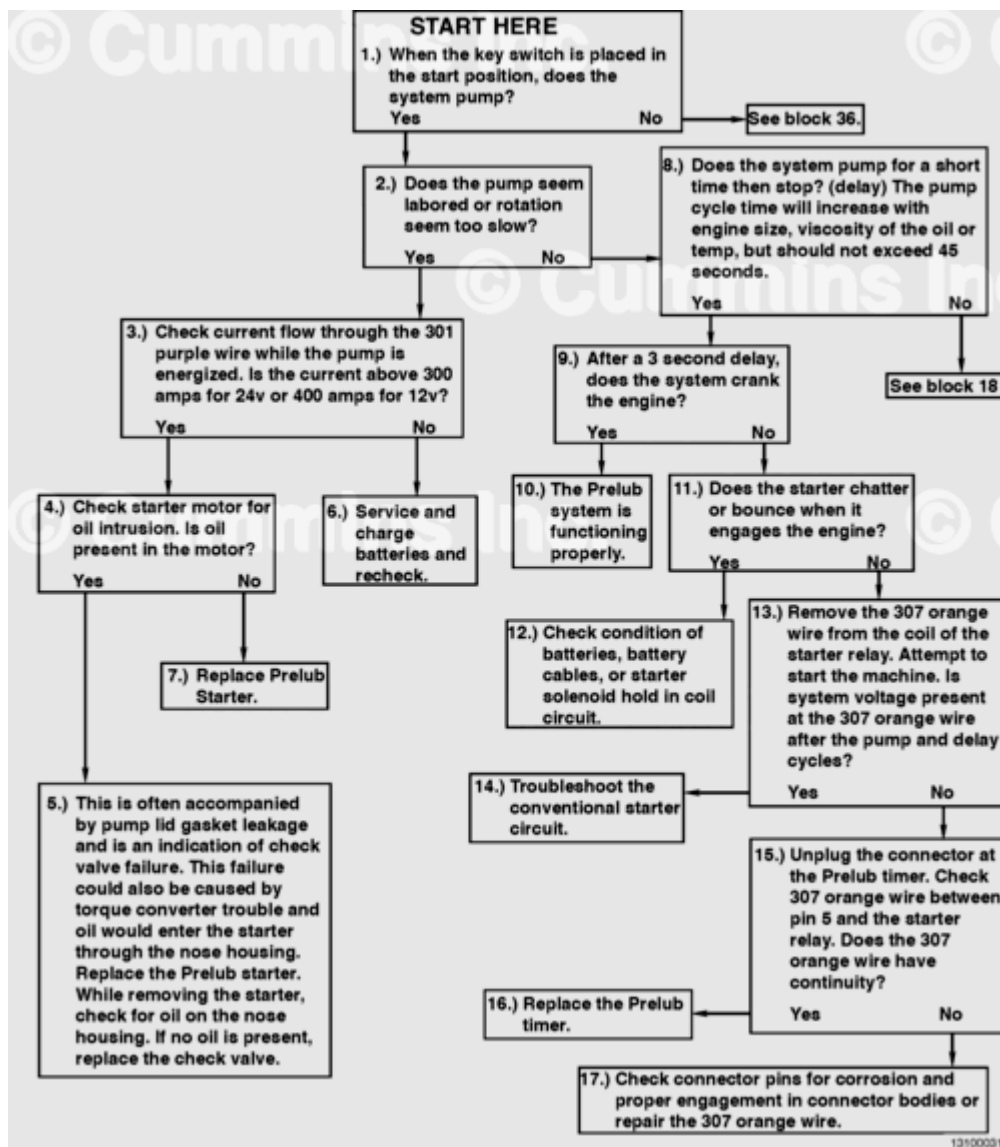
QSK19, K19, V28 and QST30 Suction Hose Installation

Note : Cummins Inc. does not recommend installing the fitting in the oil drain port of the oil pan.

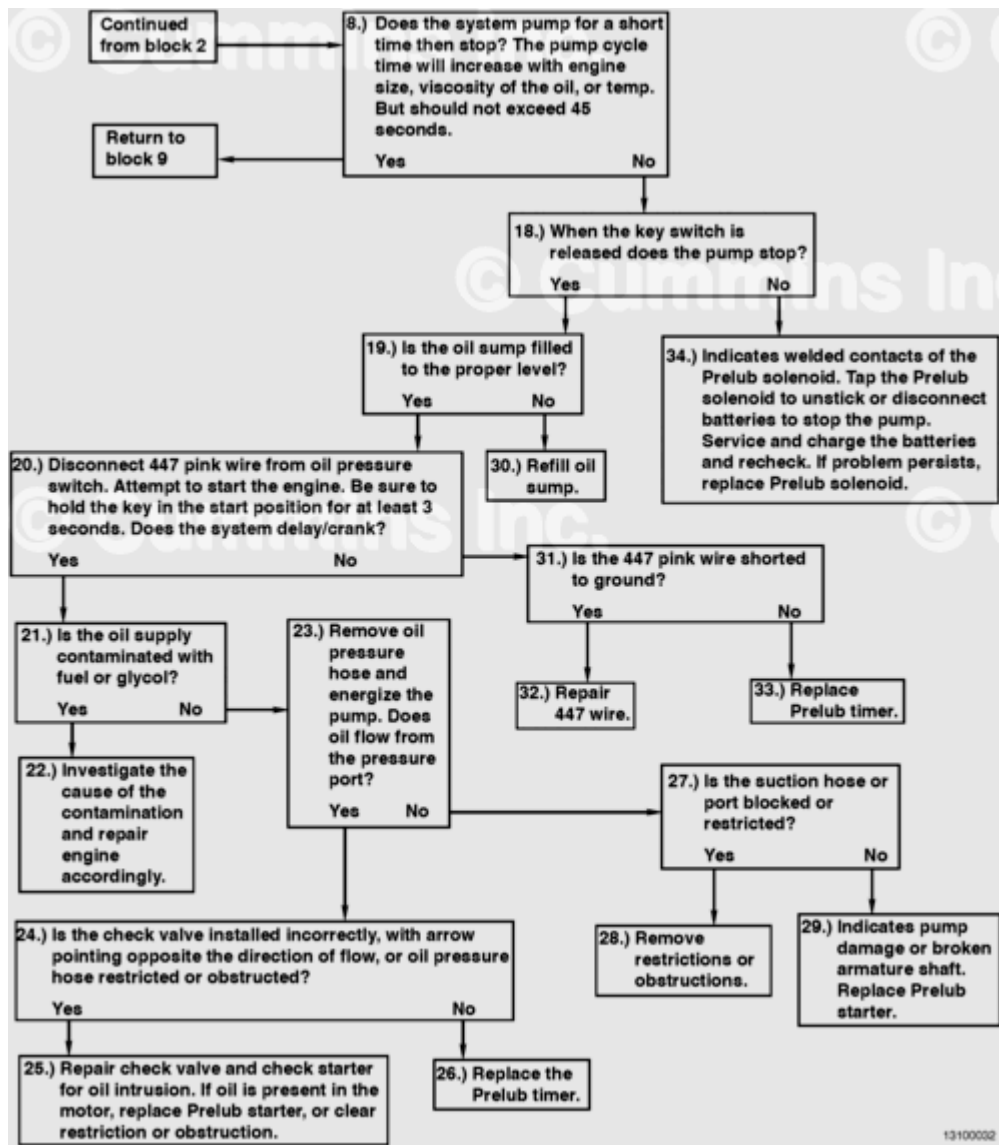
Note : The American National Standard Taper Pipe Thread (NPT) to Joint Industrial Council (JIC) fitting shown is provided in the plumbing kit.



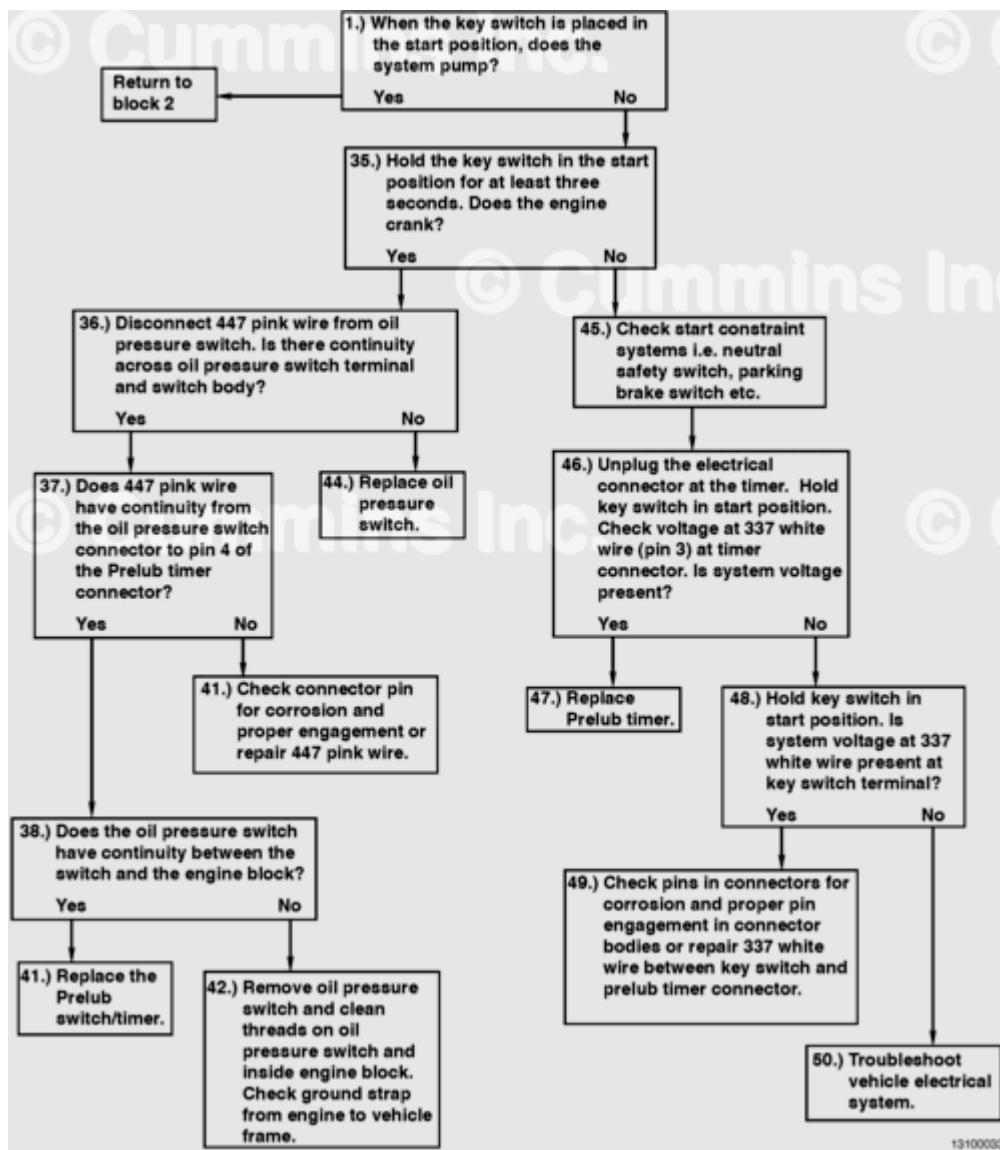
Troubleshooting Integral, Remote, and HVP Systems



Part 1.

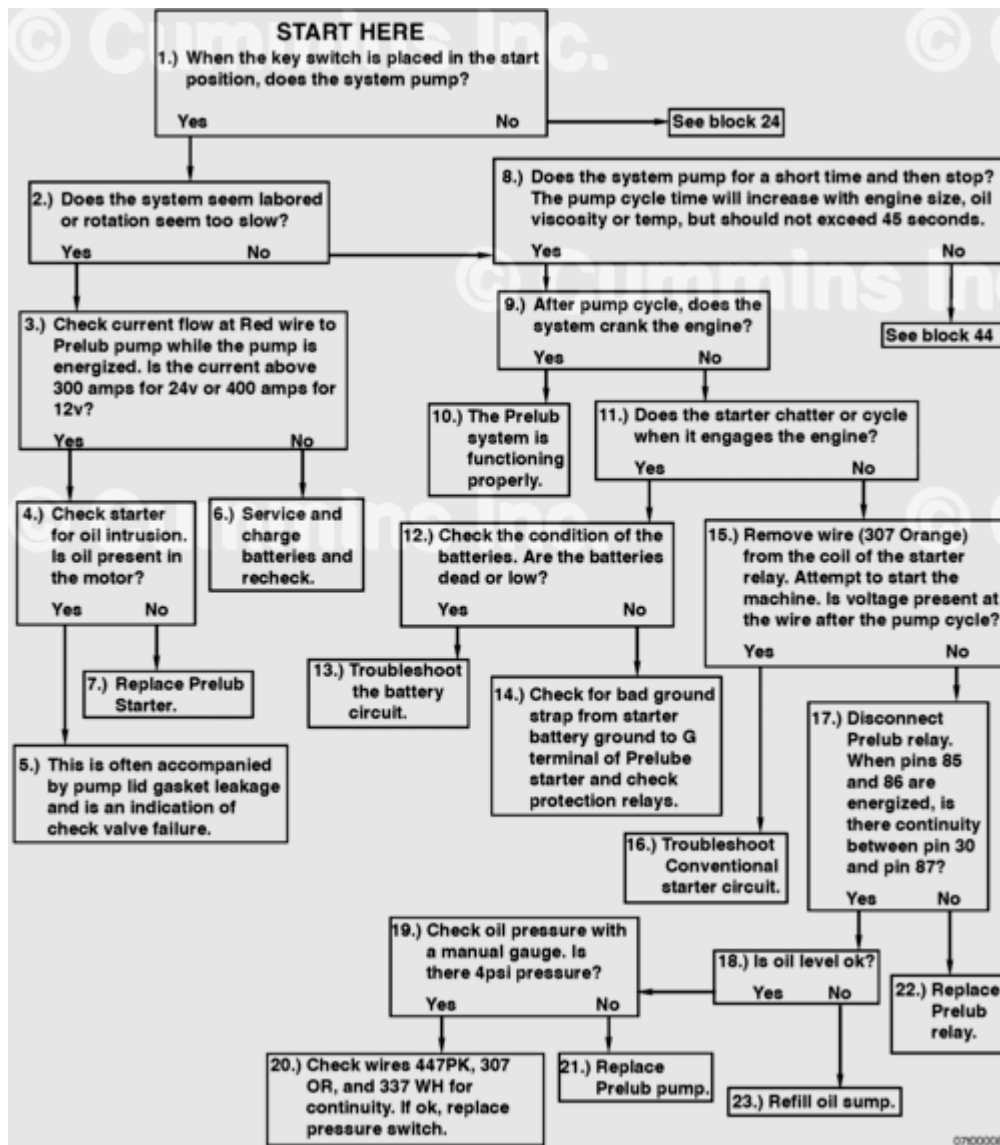


Part 2.

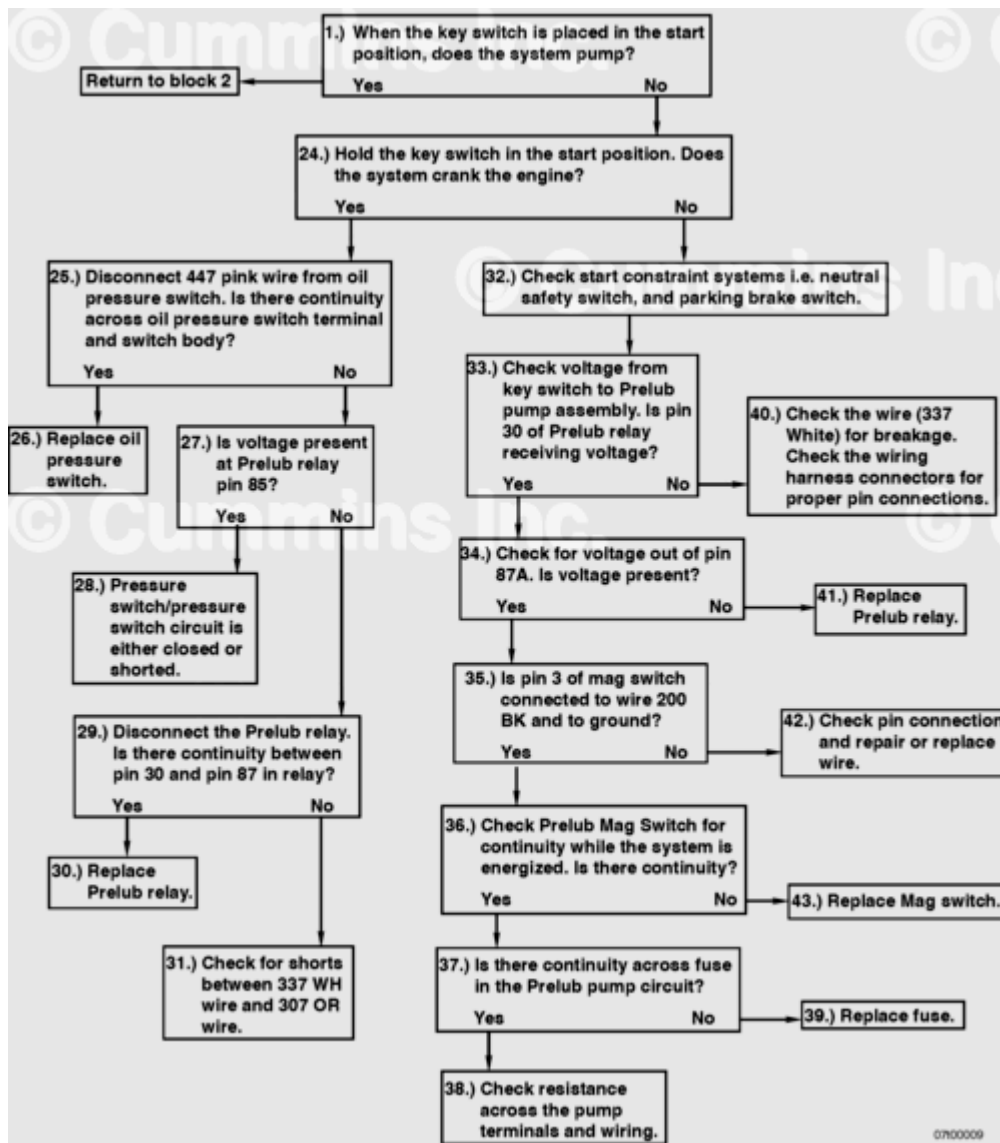


Part 3.

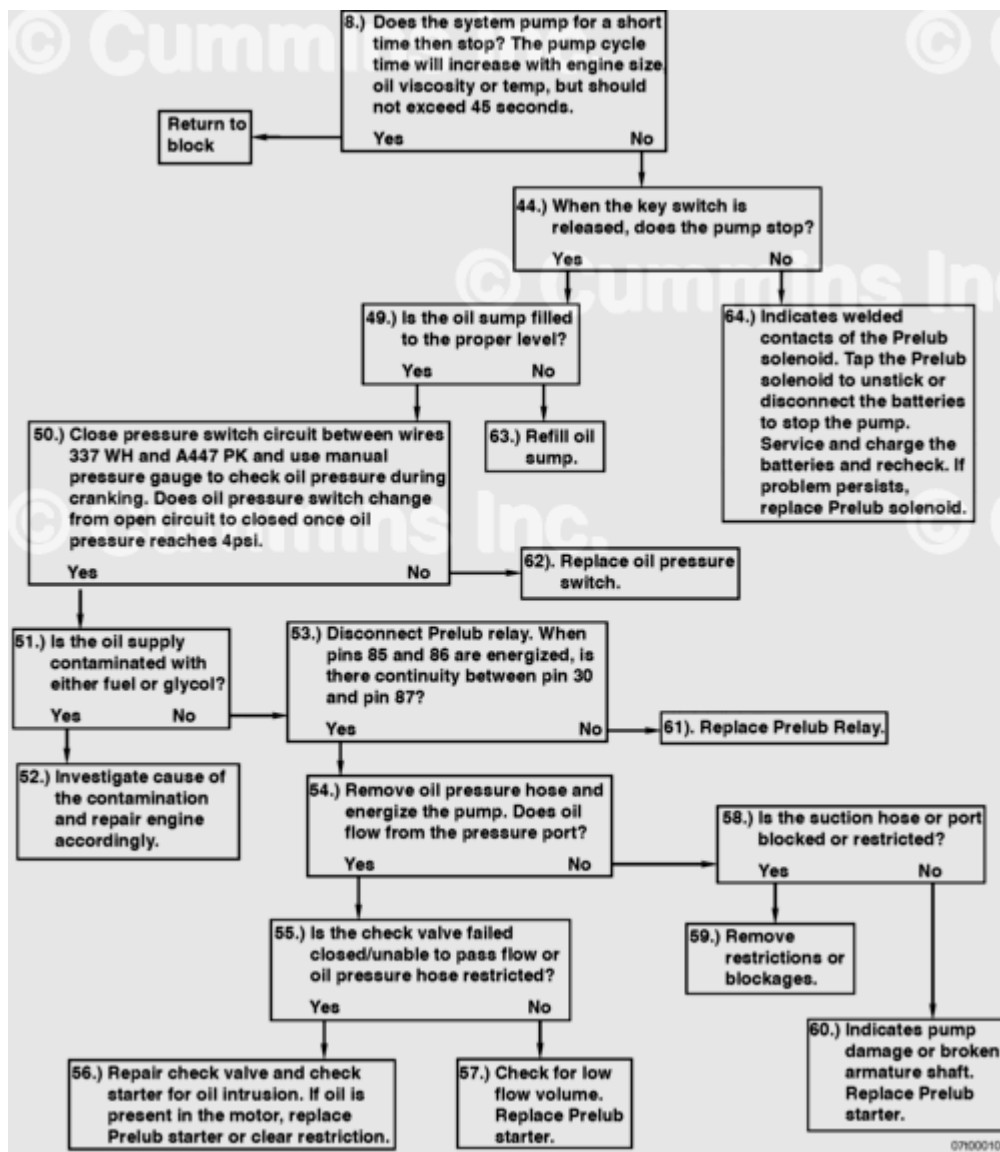
Troubleshooting Compact Prelub Pump



Part 1.



Part 2.



Part 3.

Troubleshooting the Integral, Remote, and HVP System

- PRELUB™ Phase
- A normally closed pressure valve is the key element to the PRELUB™ (pump) cycle. This switch is a ground. When no oil pressure exists and the machine ignition switch is energized, the PRELUB™ cycle will begin pumping oil until the pressure switch opens and the ground connection is broken. At this point the delay and crank phase begins.
- Delay and Crank
- Once the oil pressure switch opens (i.e., the ground connection is broken), power is automatically and **always** supplied to the delay timer where, after a 2- to 3-second delay, the starter is energized. The delay mode **always** precedes the crank mode. Thus, if the pressure switch wire (447 pink) is removed and ground is broken, the PRELUB™ unit will simply delay

and crank. The system senses “no ground connection,” assumes oil pressure has been built, and proceeds to delay and crank.

Machine Failure Modes and Suggested Troubleshooting Sequence

Starter Pumps, No Delay or Crank

- a. Indicates that ground to pressure switch (447 pink) is **not** being broken.
- b. No oil or low oil in the engine - pump can **not** build sufficient pressure to open pressure switch.
- c. Pressure switch has not closed and is holding ground.
- d. Gear pump malfunction - likely caused by debris in the pump.
- e. Oil pressure switch wire chafed and shorting to the engine block.

Starter Delays and Cranks - No Pump Mode

- a. If an operator indicates his ignition is totally inoperative, be sure he continues to hold the ignition in the crank position for 3 to 4 seconds. If the engine cranks after a short delay, this indicates a ground connection to the pressure switch has been broken. Sensing no ground (i.e., oil pressure), the PRELUB™ unit will proceed to delay and crank.
- b. Check the ground strap to the engine block. If missing, the engine block and oil pressure switch are **not** grounded.
- c. Check (447 pink) wire to pressure switch. If removed or cut, replace the wire.
- d. Check the pressure switch for failure open. Remove (447 pink) wire, then check for an open circuit between the switch terminal and the switch base. If open, simply replace the pressure switch.

Note : If the pressure switch is **not** OPEN, the machine can be started and operated until a replacement is available. The operator must simply hold the start switch for delay and crank sequence.

Starter Bounces When in the Crank Mode

- a. Check for low or dead batteries.
- b. Check the alternator output - if acceptable, replace the batteries.
- c. Check for a bad ground strap or **no** ground wire from the starter battery ground post to the G terminal of the PRELUB™ starter.
- d. Check for bad protection relays - if used in the starter circuit.

Starter Has Very Long Pump Cycle (45 Seconds) but Eventually Delays and Cranks

- a. With the exception of severe cold weather starts, PRELUB™ cycles of more than 45 seconds are **not** normal.
- b. Be sure the suction hose is 2.54 cm [1 inch] (i.e. Number 16 1-inch inside diameter (I.D.) or larger). Reducing the hose diameter will reduce pump output dramatically.
- c. Check suction-side for air leaks, loose connections, cracked fittings, or pump castings.
- d. Check the location of the oil pressure switch. It should be in a **main** oil pressure gallery. For QST30 engines, the oil pressure switch should be in the piston cooling gallery. The oil

pressure switch should **not** be tapped into a metered oil flow as in a service or governor assembly.

- e. Be sure oil of proper viscosity is being used with reference to the ambient temperature. See the engine manufacturer's specifications.
- f. Oil dilution can also cause long pump cycles. Check the dipstick for fuel or water contamination.

Starter Totally Inoperative - No Pump, Delay, or Crank Mode

- a. If the starter is totally inoperative, this indicates a voltage interruption in the system.
- b. Check the voltage from the keyswitch to the PRELUB™ switch assembly, and check the wire (337 white) for breakage. Check the wiring harness connectors for proper pin connections, and check transmission controls and neutral start constraints.
- c. If the system voltage is present at the wire (337 white) and no pump is activated when the pressure switch is grounded, a failure is indicated in the PRELUB™ switch timer. Replace the timer.

Starter Pumps, Delays, Then Does Not Crank After 2- to 3-Second Delay

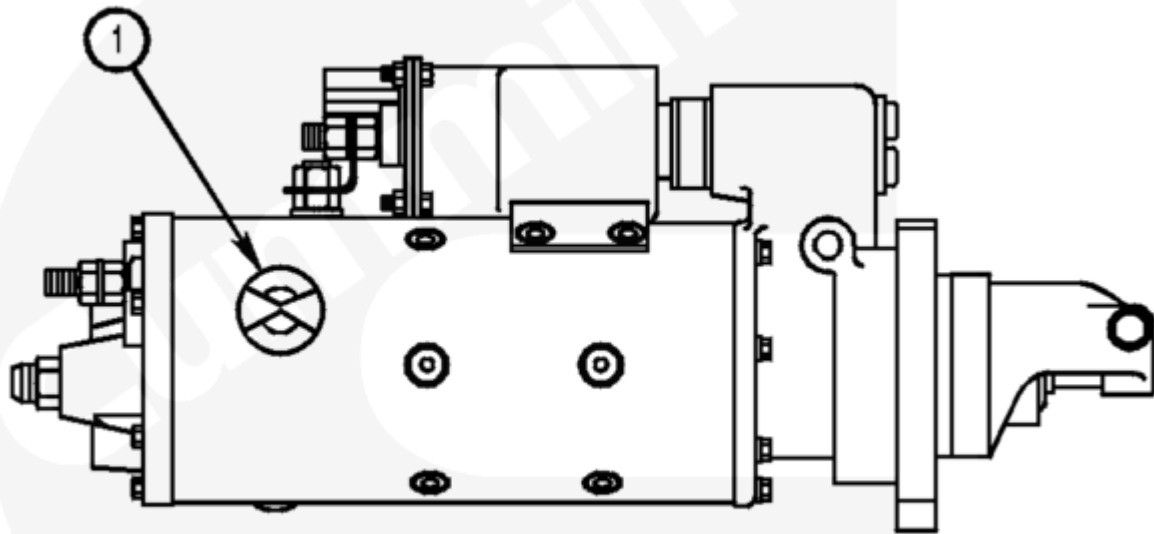
- This indicates either a timer failure or starter circuit problem.
- Remove the wire (307 orange) from the starter relay and test with a voltage meter for system voltage signal (i.e., 24-VDC) after a 2- to 3-second delay. If the wire (307 orange) presents system voltage after a 2- to 3-second delay, check the starter relay and starter motor as for a conventional starter. If system voltage is **not** present, first check the connections within sure seal connectors at pin position 5 of the main harness and pin position 2 of the two wire connector (307 orange). If **no** voltage is present and **no** connector problems are found, replace the PRELUB™ timer.

On Dual-Starter System - PRELUB™ Starter Pumps While Conventional Starter Cranks Continuously

- Check to make sure the second starter is wired to its own starter relay.

Harness or Switch-Timer Shows Signs of Overheating

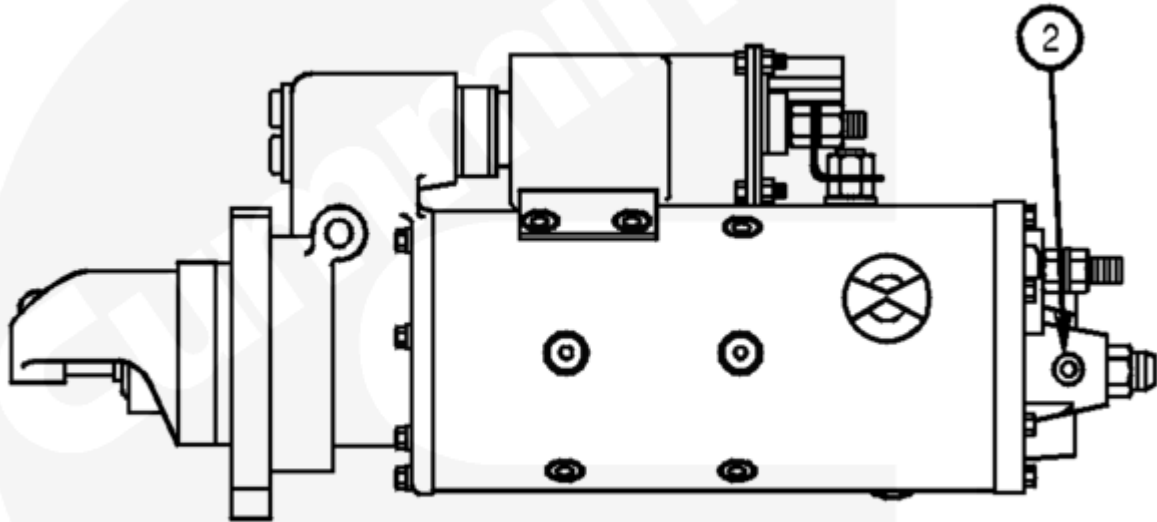
- This is an Indication of Motor Current Being Too High. Check as Follows:
- Make sure that all electrical cables and connections are in good condition and are tightened to the proper torque.
- Check the machine batteries to confirm that they are properly charged according to the OEM's recommendations.
- Measure the amperage draw of the motor while it is in the PRELUB™ cycle. If the current draw of the motor is more than 300 amps (24-VDC) during the PRELUB™ cycle, check for the presence of oil in the motor cavity. Also, confirm proper operation of the check valve assembly. Perform these checks as follows:



13100034

1. For safety, disconnect electrical power from the PRELUB™ starting system.
2. Remove the inspection plug (1) from the frame of the PRELUB™ starting motor.
3. Carefully inspect the motor cavity with a light for the presence of engine oil. Oil can be present either in large amounts or as a light mist coating on internal components. Oil in the motor cavity in any quantity indicates a malfunctioned motor.

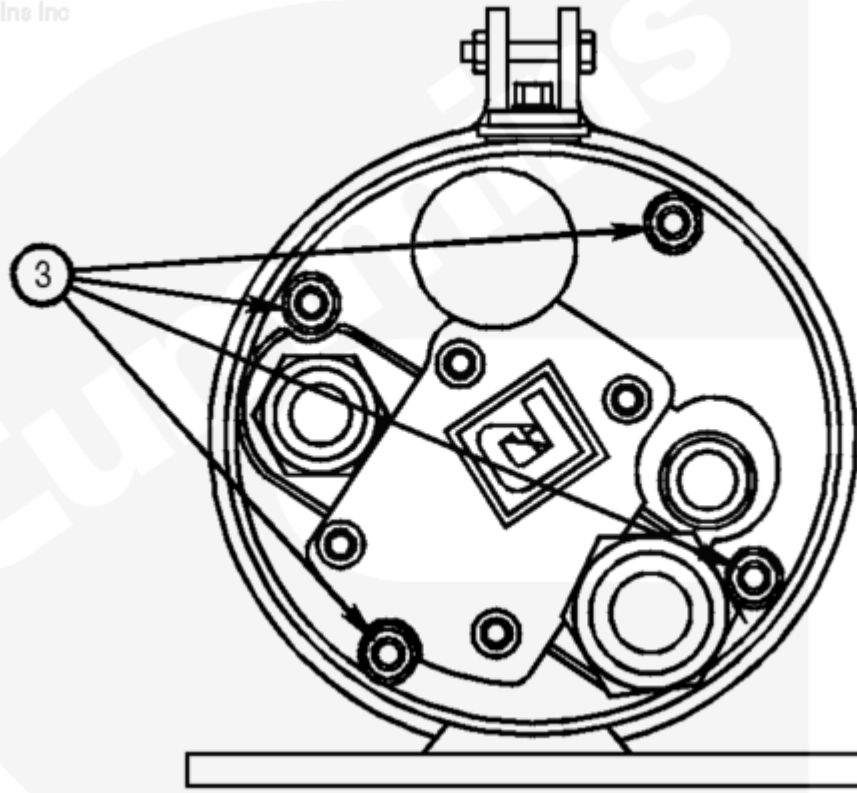
Note : Brush dust will be present in all motors. Do not confuse brush dust with oil. Brush dust will appear as a black/brown dry coating within the motor.



13100035

1. If oil is present in the motor, install a 1.38-mPa [200-psi] pressure gauge in the pump side port (2). Start the engine, and observe the pressure gauge. If there is **any** indication of pressure, replace the PRELUB™ starting motor and check valve. If there is **no** pressure indicated, replace the PRELUB™ starter.
2. If **no** oil is present, carefully install the inspection plug.

41RM

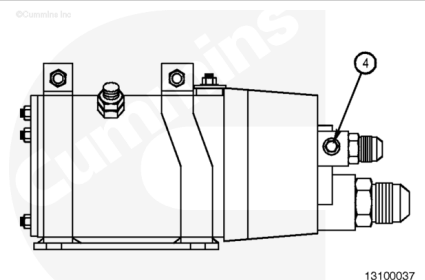


13100036

1. For safety, disconnect electrical power from the PRELUB™ starting system.
2. Remove the pump housing bolt that is closest to the 6-o'clock position on the PRELUB™ remote unit (3).
3. Inspect the shank of the pump housing bolt and the bolt hole for the presence of oil. Oil can be present either in large amounts or as a light mist on the bolt shank or in the bolt hole. Oil in the motor cavity in any quantity indicates a malfunctioned motor.

Note : Brush dust will be present in all motors. Do not confuse brush dust with oil. Brush dust will appear as a black/brown dry coating within the motor.

1. If oil is present in the motor, install a 1.38-mPa [200-psi] pressure gauge in the pump side port (4). Start the engine, and observe the pressure gauge. If there is **any** indication of pressure, replace the PRELUB™ remote unit and check valve. If there is **no** pressure indicated, replace the PRELUB™ remote unit.
2. If **no** oil is present, carefully install the pump housing bolt.



13100037

Torque Value: 9.6 n·m [85 in-lb]

Troubleshooting the No Delay Compact PRELUB™ Switching System

PRELUB™ Phase

The key difference of the Compact PRELUB™ system is the use of a normally **open** oil pressure switch and the elimination of the timer which is replaced by the PRELUB™ relay. When no oil pressure exists and the machine ignition switch is energized the PRELUB™ cycle will begin pumping oil until the pressure switch closes. At this point the crank phase begins.

Crank Phase

Once the oil pressure switch closes the PRELUB™ relay is energized switching power to the starter motor. There is **not** a timer in the Compact PRELUB™ system meaning the system will crank immediately after building oil pressure and closing the pressure switch.

Machine Failure Modes and Suggested Troubleshooting Sequence

1. Starter pumps, No Crank
 - a. Pressure switch has failed open
 - b. Low Oil - insufficient pressure to close switch
 - c. Gear pump failure
 - d. Check 447 pink wire to pressure switch. If this wire is removed or broken, replace the wire
 - e. Prelub relay is **not** switching
2. Starter Cranks - No Pump Mode
 - a. Oil pressure switch shorted
 - b. Check pressure switch for failure closed
 - c. Short in pressure switch circuit. Prelub Relay stuck at pin 87
3. Starter cycles or chatters when in crank mode.
 - a. Check for low or dead batteries
 - b. Check for a bad ground strap or no ground wire from the starter battery ground to the G terminal of the PRELUB™ starter.
 - c. Check for bad protection relays.
4. Starter has very long pump cycle (45+ seconds) but eventually cranks. With the exception of severe cold starts, PRELUB™ cycles of more than 45 seconds are **not** normal.
 - a. Check suction hose has 2.54 cm [1 inch] inner diameter (i.e. Number 16 1-inch inside diameter (I.D.) or larger). Reducing the hose diameter will reduce pump output dramatically
 - b. Check suction side for air leaks, loose connections, cracked fittings, or pump castings
 - c. Check location of oil pressure switch - should be in main oil pressure gallery
 - d. Be sure oil of proper viscosity is being used with reference to the ambient temperature. See engine manufacturer's specifications.
 - e. Oil dilution can cause long pump cycles - Check dipstick for fuel or water contamination
 - f. Remove oil filter and check for metal debris
5. Starter totally inoperative - No pump or crank mode. If the starter is totally inoperative, this indicates a voltage interruption in the system.
 - a. Check voltage from key switch to prelubrication pump assembly, and check the wire (337 White) for breakage. Check the wiring harness connectors for proper pin connections, and check transmission controls and neutral start constraints
 - b. If switch voltage is present at the wire (337 White) check voltage is present at wire (337 White) at pin 30
 - c. Check voltage to pin 87A

- d. Check voltage to pin 4 of the magnetic switch connector
 - e. Check pin 3 to ground
 - f. Check magnetic switch
 - g. Check fuse in prelubrication motor circuit
 - h. Check PRELUB™ pump resistance and earth
6. On dual starter system - prelubrication starter pumps while conventional starter cranks continuously.
- a. Check to make sure second starter is wired to its own starter relay
7. Harness shows signs of overheating.
- a. Indication of motor current being too high.

Warranty

Coverage

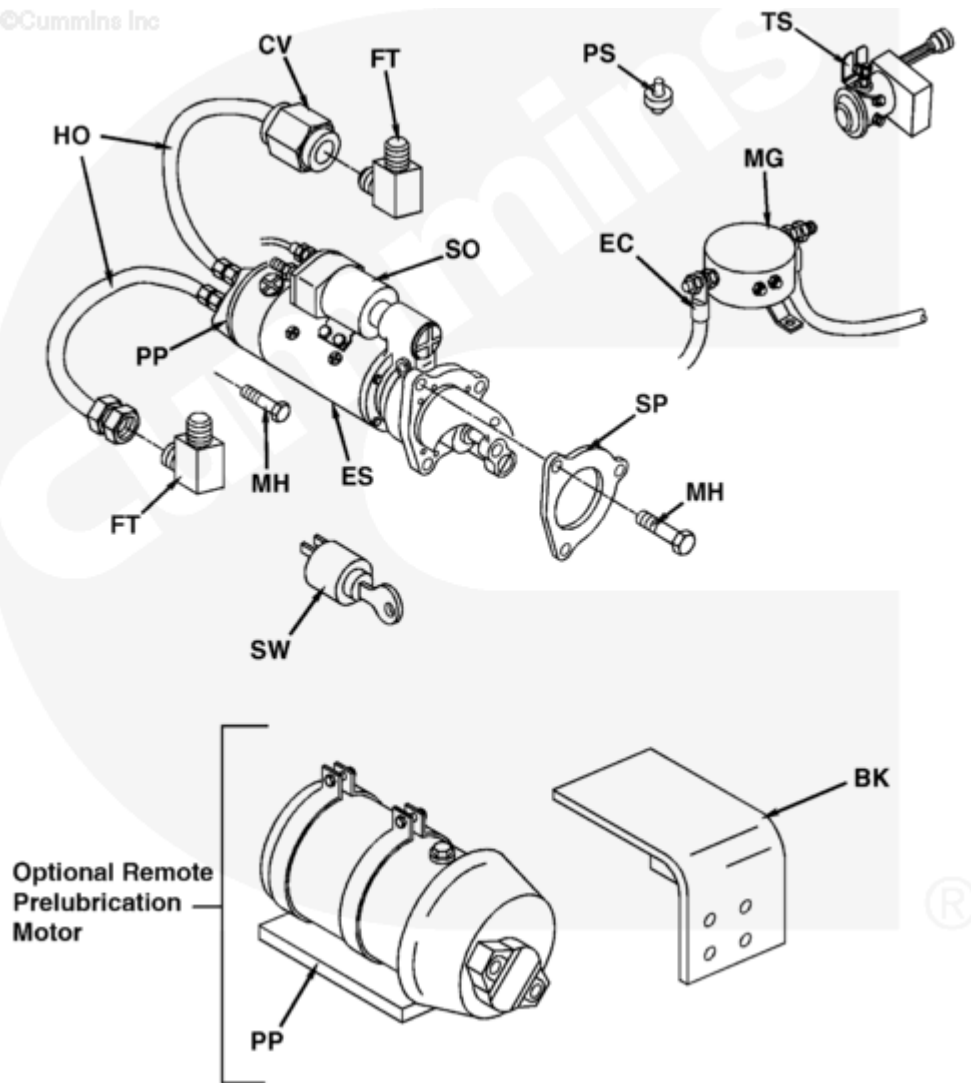
Cummins® prelubrication system carries Cummins® Industrial and Marine Warranty if purchased with the engine or installed on the engine prior to date in service. If the system is purchased through parts the distribution center (PDC) for retrofit purposes, Cummins® New Parts Warranty is applicable.

Cummins® Industrial Warranty coverages are described in Bulletin Number 3381321, revised 1/96. Cummins® Marine Warranty coverages are described in Bulletin Number 3381775 revised 8/98. Cummins® New Parts Warranty is a 1-year, 3600-hour parts and workmanship coverage as described in the Warranty Bulletin, Bulletin Number 3381213. In the event of future changes to either coverage, the product will carry the applicable warranty present at the time of the purchase.

Claims

Administrative procedures involved in filing claims for this product are described in the Cummins® Warranty Administration Manual. Account code 67 **must** be specified for New Engine Warranty and account code 62 **must** be specified for New Parts Warranty.

Warranty Failure Codes



PM Prelubrication Motor

Mode of Failure

(BK) Bracket	BE BR CU MA MM
(CV) Check Valve	BR CO MA MM WP
(EC) Electrical Connection	BR CO MA SR WP
(ES) Starting Motor	BR CO ID MA MM SR WP
(FT) Fitting	BR CO MA MM WP
(HO) Hose	BR CU LK MA WP
(MG) Magnetic Switch	BR CO MA SR WP
(MH) Mounting Hardware	BE BR CU MA MM WP
(PP) Prelubrication Pump	BR CO DR LO MA MM WP
(PS) Oil Pressure Switch	BR CO MA SR WP
(SO) Starter Solenoid	BR CO MA SR WP
(SP) Spacer	BR CU MA MM WP
(SW) Switches	BR CU ID MA SR WP
(TS) Timer Solenoid	BR CO MA SR WP

Mode of Failure	
(WH) Wiring Harness	BR CO MA SR WP

Standard Repair Times (SRT)			Remove d From Chassis	In-chassis Service Accessibility Codes			
Procedure Number and Description			R	A	B	C	D
00-659	01	Includes: Verify Problem:	0.2	0.2	0.2	0.3	0.3
		•Check the engine oil level					
		•Check for chaffed or shorted wire to pressure switch					
		Includes:					

Standard Repair Times (SRT)			Remove d From Chassis	In-chassis Service Accessibility Codes			
Procedure Number and Description			R	A	B	C	D
	02	Perform checks in step 01:	0.2	0.2	0.2	0.3	0.4
		•Install the oil pressure gauge in the camshaft rifle					
		•Operate the prelubrication motor.					
		•Check for oil pressure					
		•Check oil level					
		NOTE: Referenc e the following repair procedur es and times as necessar y:					
		Pressure Switch - Replace					
		Prelubric ation Pump - Replace					

Standard Repair Times (SRT)			Remove d From Chassis	In-chassis Service Accessibility Codes			
Procedure Number and Description			R	A	B	C	D
00-660	01	Troubles hoot - Prelubric ation Starter Motor Prelubric ates Continuo usly Regardle ss of Keyswite h Position					
		Includes: Verify Problem:	0.1	0.1	0.1	0.1	0.1
		•Disconn ect wire					

Standard Repair Times (SRT)			Remove d From Chassis	In-chassis Service Accessibility Codes			
Procedure Number and Description			R	A	B	C	D
	02	Perform checks in step 01:	0.3	0.3	0.4	0.5	0.6
		Replace timer solenoid if disconne cting wire prevents continou s operatio n of the prelubric ation pump					
		Bypass prelubric ation system if replacem ent timer solenoid available	0.3	0.3	0.4	0.5	0.6
		NOTE: Referenc e the following repair procedur es					
		•Timer Solenoid - Replace					

Standard Repair Times (SRT)			Remove d From Chassis	In-chassis Service Accessibility Codes			
Procedure Number and Description			R	A	B	C	D
00-661		Troubles hoot - Prelubric ation Starter Delays and Crank Only					

Standard Repair Times (SRT)						
Remove and Install Procedures	Procedure Number	Removed from chassis R	A	B	C	D
Prelubricat ion Starter Motor/Pu mp	13-125	0.3	0.5	0.6	0.7	0.8
Prelubricat ion Timer Solenoid	15-134	0.1	0.2	0.3	0.4	0.5
Prelubricat ion Oil Pressure Switch	15-135	0.1	0.1	0.2	0.3	0.3
Prelubricat ion Wiring Harness	15-136	0.2	0.3	0.3	0.4	0.5

Standard Repair Times (SRT)			Remove d From Chassis	In-chassis Service Accessiblity Codes			
Procedure Number and Description			R	A	B	C	D
00-662	01	Troubleshoot - Prelubrication Starter Motor Delays and Cranks					
		Includes: Verify problem:					

Standard Repair Times (SRT)			Remove d From Chassis	In-chassis Service Accessibility Codes			
Procedure Number and Description			R	A	B	C	D
		•Check the engine block ground	0.3	0.3	0.4	0.5	0.6
		•Check wire to pressure switch					
		•Remove wire to pressure switch					
		•Check for open circuit between switch and terminal and switch base					
		NOTE: Referenc e the following repair procedur es and times as necessar y					
		•Pressur e switch - Replace					

Standard Repair Times (SRT)			Remove d From Chassis	In-chassis Service Accessibility Codes			
Procedure Number and Description			R	A	B	C	D
		Troubles hoot - Prelubric ation Starting Circuit is Irregular When Cranking Mode Includes: Verify Problem:					

Standard Repair Times (SRT)			Remove d From Chassis	In-chassis Service Accessibility Codes			
Procedure Number and Description			R	A	B	C	D
00-662	01	•Check for low or dead batteries	0.2	0.3	0.4	0.5	0.6
		•Check for alternato r output					
		•Check for broken wire between battery ground and G terminal of starter solenoid					
		•Check for bad protectio n relays if used in starting system					
		NOTE: Referenc e the following repair procedur es and times as necessar y					
		•Alternat or - Replace					

Standard Repair Times (SRT)			Remove d From Chassis	In-chassis Service Accessiblity Codes			
Procedure Number and Description			R	A	B	C	D
		•Batterie s - Replace					
		•Relay(s) - Replace					

Standard Repair Times (SRT)			Remove d From Chassis	In-chassis Service Accessibility Codes			
Procedure Number and Description			R	A	B	C	D
00-663	01	Prelubric ation Starter Motor - has long prelubric ation cycle but delays and cranks Includes: Verify Problem:					
		•Check oil level	0.4	0.4	0.5	0.6	0.7
		•Check suction hose for proper size and plugging					
		•Check for air leaks in suction side					
		•Check location of oil pressure switch in camshaf t rifle					
		•Check for proper viscosity of oil					

Standard Repair Times (SRT)			Remove d From Chassis	In-chassis Service Accessibility Codes			
Procedure Number and Description			R	A	B	C	D
00-664	01	Prelubric ation Starter Motor - has no prelubric ation, no delay, no crank Includes: Verify Problem:					
		•Remove wire to pressure switch	0.1	0.1	0.1	0.1	0.1
		•Activate machine starter switch for several seconds					
		•Replace timer solenoid if starter delays and cranks					

Standard Repair Times (SRT)			Remove d From Chassis	In-chassis Service Accessibility Codes			
Procedure Number and Description			R	A	B	C	D
		•Measure e voltage from machine starter switch to timer solenoid when activated	0.2	0.2	0.2	0.3	0.3
		NOTE: Referenc e the following repair procedur es and times as necessar y					
		•Timer Solenoid - Repalce					

Standard Repair Times		Remove d from chassis	In-chassis Service Accessibility Codes			
Procedure Number and Description		R	A	B	C	D
	Troubleshoot - Prelubrication Starter Motor Prelubricates, Delays, Then Does Not Crank					
	Includes: Verify problem:					

Standard Repair Times			Remove d from chassis	In-chassis Service Accessibility Codes			
Procedure Number and Description			R	A	B	C	D
00-665	01	•Place jumper wire between the S-post connection of the solenoid on the prelubrication starter and the ignition switch wire, and activate switch.	0.2	0.2	0.2	0.3	0.3
		•Replace the timer solenoid if jumper wire cranks engine					
		•Replace solenoid or prelubrication starter or pinion drive if jumper wire fails to crank engine					

Standard Repair Times			Remove d from chassis	In-chassis Service Accessibility Codes			
Procedure Number and Description			R	A	B	C	D
		NOTE: Referenc e the following repair procedur es and times as necessar y					
		•Timer Solenoid - Replace					
		•Starter Solenoid - Replace					
00-666		Troubleshoot - Conventional Starter Engages Flywheel During the Prelubrication Cycle					
		Includes: Verify problem:					
	01	•Check wiring with respect to S-post connecti on on conventi onal starter.	0.1	0.1	0.1	0.2	0.2

Document History

Date	Details
2011-07-06	Module Created
2011-7-6	Include additional prelube systems.

Date	Details
2011-11-2	QSOL Quick Fix Reason: QSOL Display Problem Notes: none
2011-11-2	QSOL Quick Fix Reason: Outfile Notes: none
2011-12-7	QSOL Quick Fix Reason: QSOL Display Problem Notes: none

Last Modified: 07-Dec-2011
